Orientation

What we have seen:

- RLMPC, why does it work & different flavors

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- RLMPC, why does it work & different flavors

What we will do now:

- The theory is not about MPC, it is about MDPs (MPC is a special case)
- It has broader implications for AI and model-based decision making
- Provide (tentative) practical ideas for in Sim2Real

AI for Operational Decisions
Some perspectives from recent research

Prof. Sebastien Gros

Department of Cybernetic
Faculty of Information Technology
NTNU

Freiburg PhD School

Outline

1 Decisions from data

2 AI models for Decision



Model-Free Pathway - "Pure" Reinforcement Learning



Actions

State

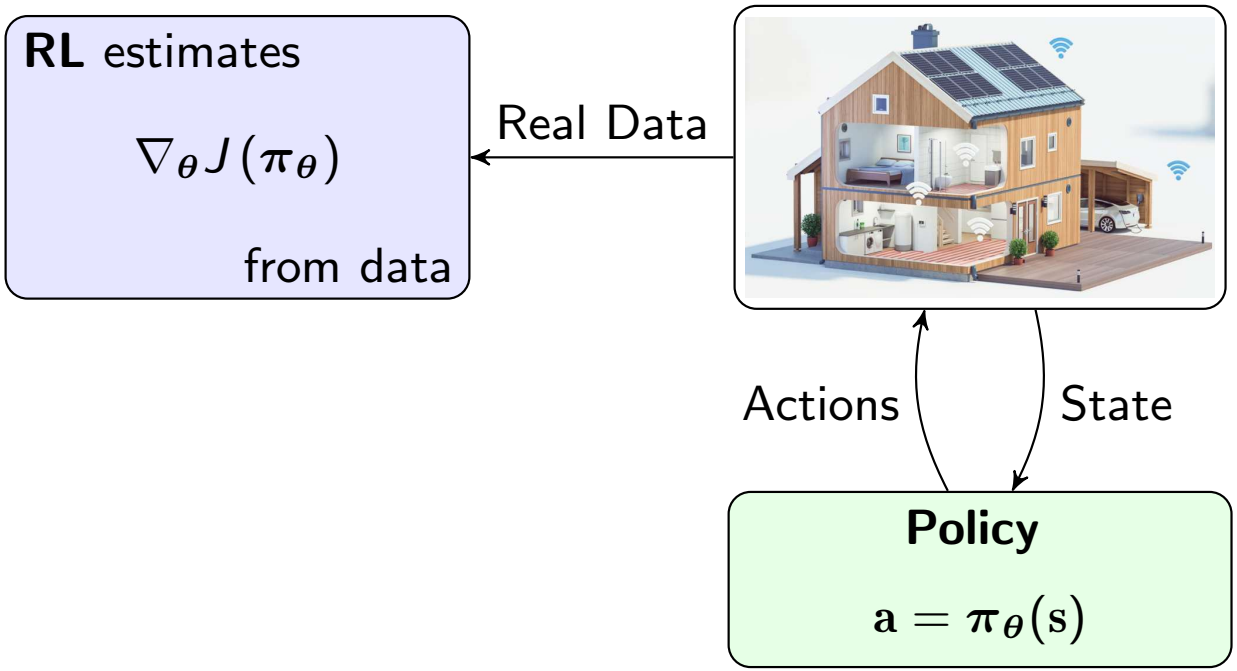
Policy

$$\mathbf{a} = \pi_\theta(\mathbf{s})$$

- Define policy π_θ as a parametrized function
- Optimize policy as:

$$\min_\theta \; J(\pi_\theta)$$

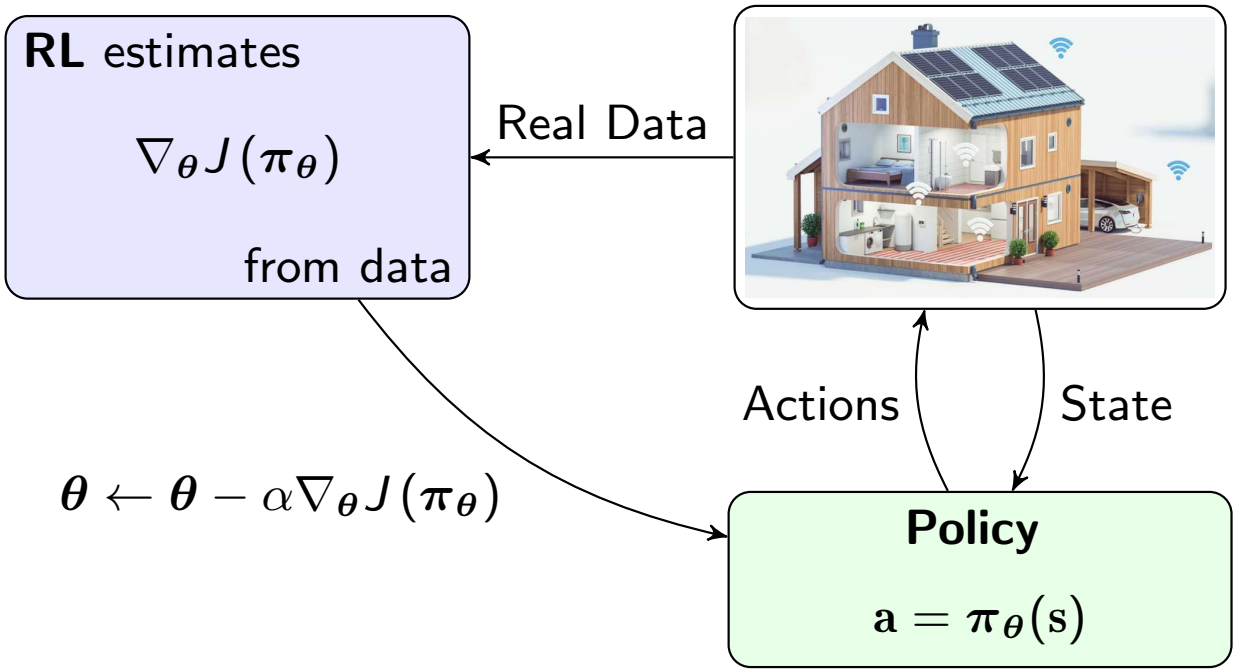
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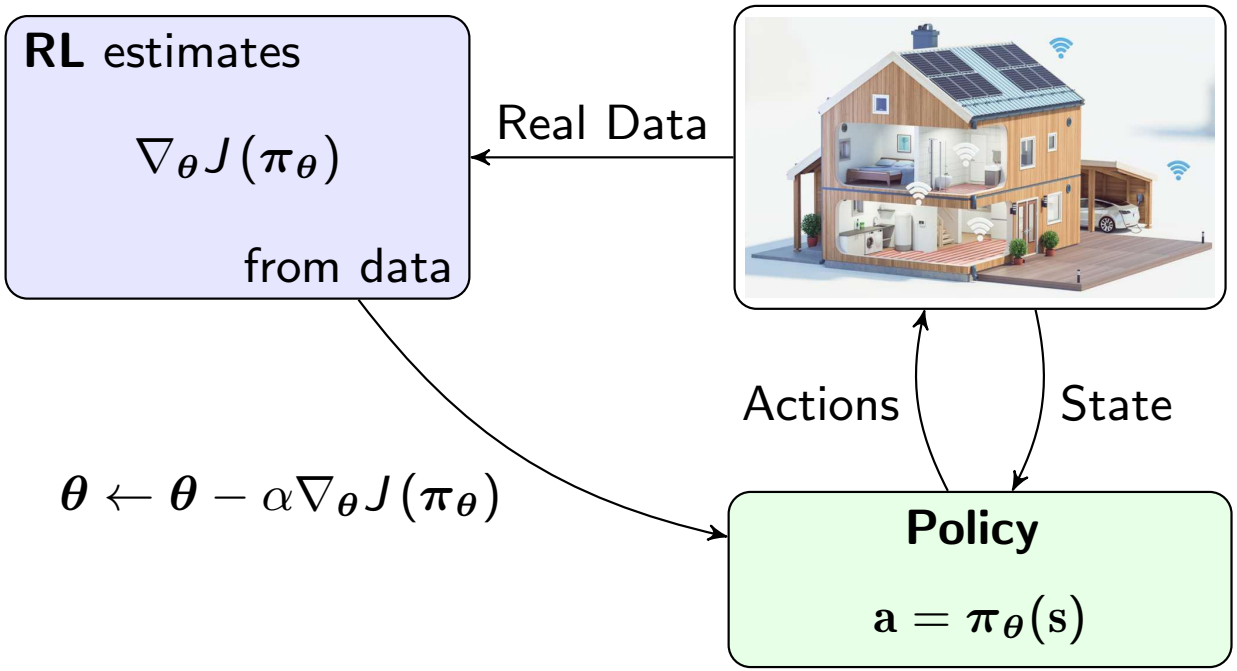


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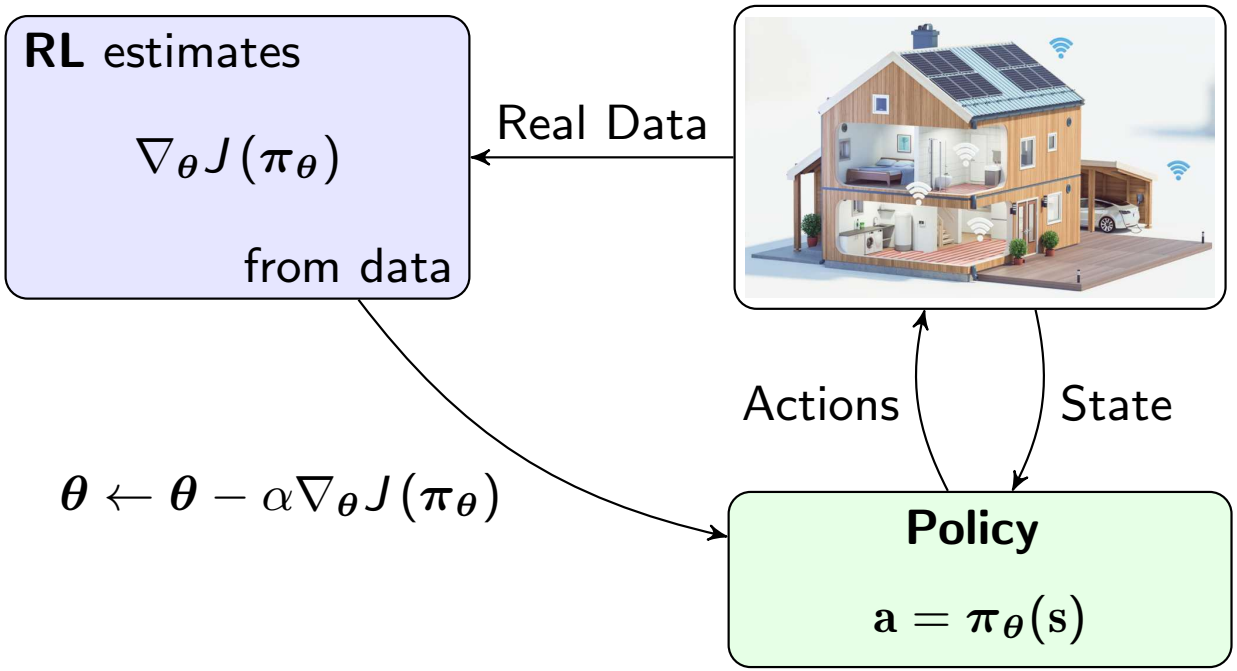
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Remarks

- $\pi_\theta(\mathbf{s})$ often from DNN
- End-to-end training requires a lot of data and effort
- Difficult to provide explainability and guarantees on the policy we obtain
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$\rightarrow$policy is often trained on **high-fidelity simulations**: embed knowledge, reduce costs, manage safety, promote explainability, control the training

Model-based Pathway - Predictive AI Models (data-driven, ML, etc)

Real world: $\mathbf{s}_{k+1} \sim \rho(\cdot \,|\, \mathbf{s}_k, \mathbf{a}_k)$

- Data with enough "richness"

Model-based Pathway - Predictive AI Models (data-driven, ML, etc)

One-step model

AI model e.g.

$$\hat{\mathbf{s}}_{k+1} \sim \rho_{\boldsymbol{\theta}}(\cdot \,|\, \mathbf{s}_k, \mathbf{a}_k)$$

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Model parameters $\boldsymbol{\theta}$

- Selected such that model $\rho_{\boldsymbol{\theta}}$ "ressembles" reality ρ, using data (+ physics)
- Deterministic models are a special case of $\rho_{\boldsymbol{\theta}}$
- Many methods, from Least Squares and MLE to Bayesian and Adversarial Learning
- Still, in most cases $\rho_{\boldsymbol{\theta}}$ "simplifies" ρ because
  - need for huge amounts of data to decide $\boldsymbol{\theta}$ if model is very rich
  - computationally demanding simulations if $\rho_{\boldsymbol{\theta}}$ is expensive to sample from
- Then
  - biases in case of insufficiently rich model structure
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  - only few first moments are correct (typ. mean + variance)

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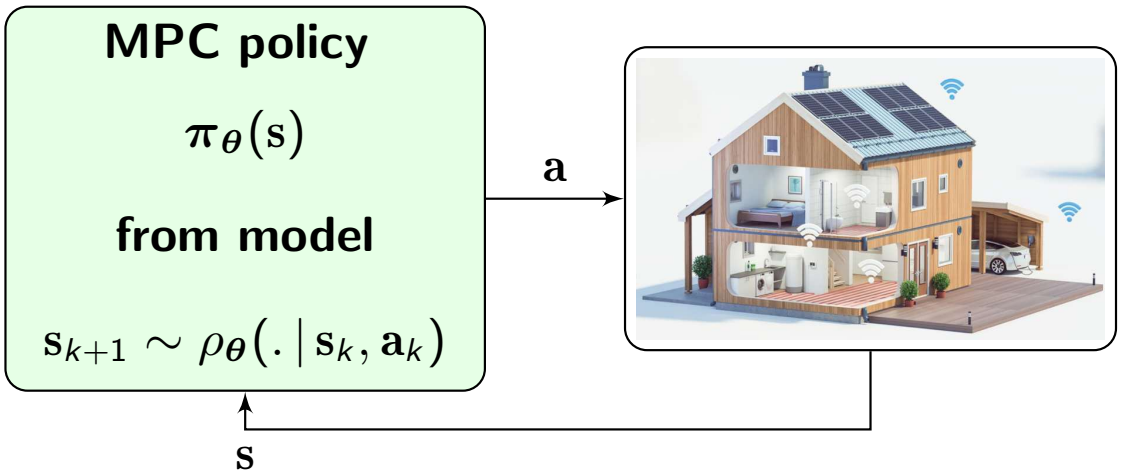
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Back to MPC - A "Classical" View on Decision Making

Policy $\pi_\theta(\mathrm{s})$ given implicitly by

$$
\begin{aligned}
\min_{\mathbf{x},\mathbf{u}} \quad & \mathbb{E}\left[T\left(\mathbf{x}_N\right)+\sum_{k=0}^{N-1} L\left(\mathbf{x}_k,\mathbf{u}_k\right)\right] \\
\text{s.t.} \quad & \mathbf{x}_{k+1} \sim \rho_{\boldsymbol{\theta}}\left(\mathbf{x}_k,\mathbf{u}_k\right) \\
& \mathbf{x}_0 = \mathbf{s}
\end{aligned}
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use $\pi_\theta(\mathrm{s}) = \mathbf{u}_0^\star$ on the system

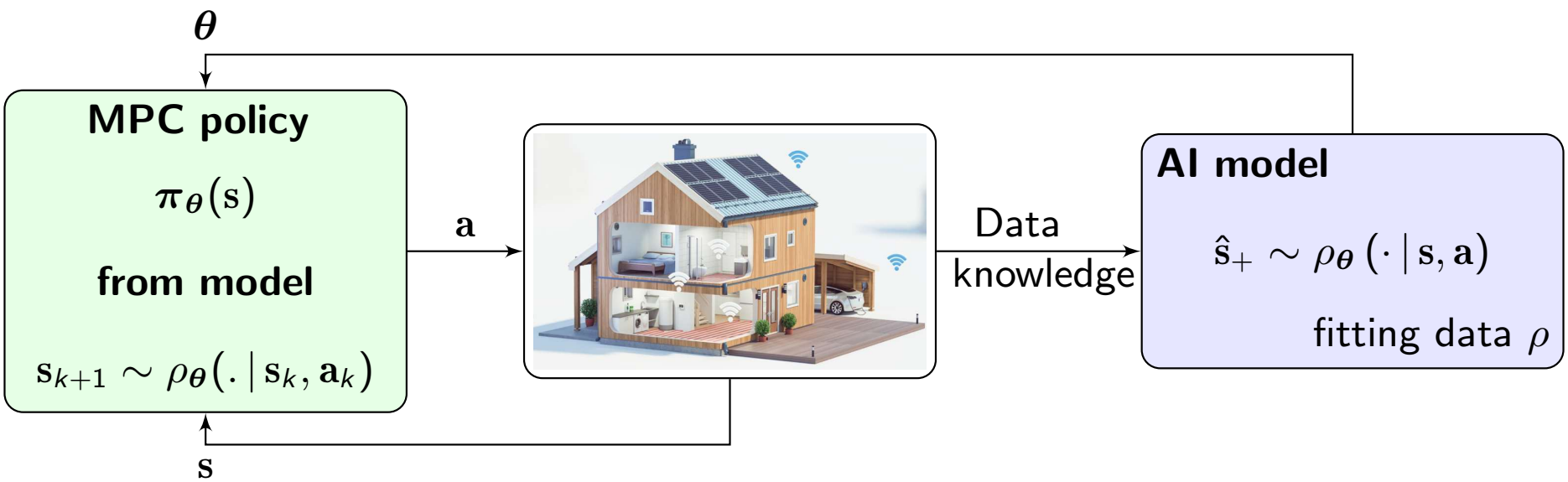


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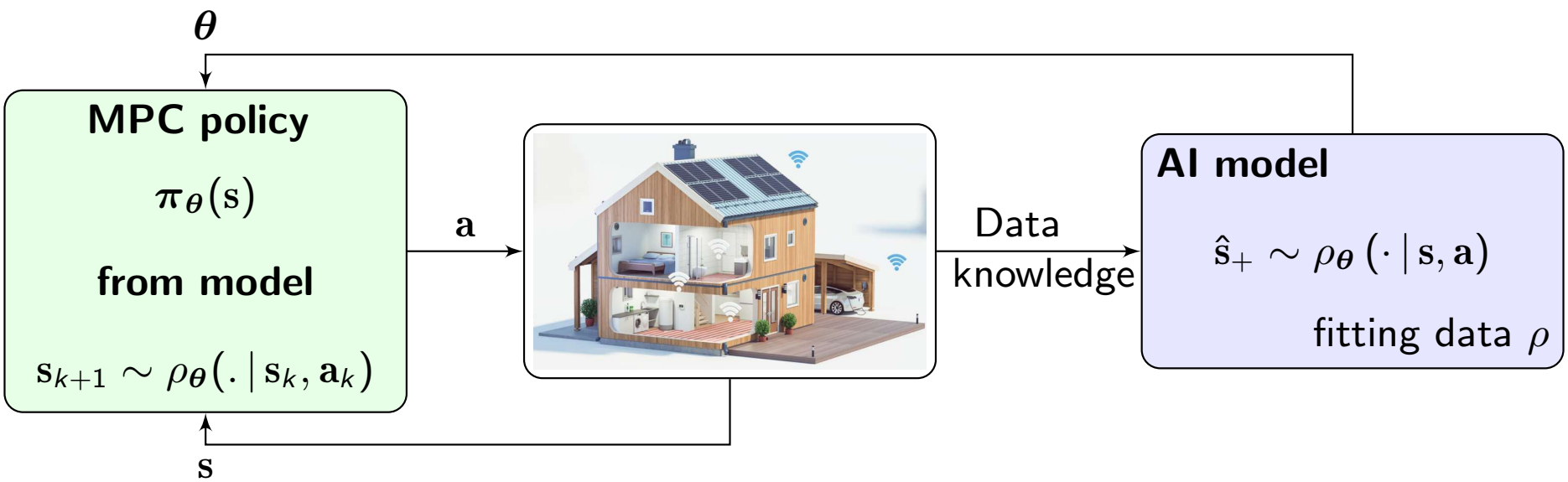
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Defines a paradigm...

- Performance from model accuracy

  i.e. $\rho_{\boldsymbol{\theta}}$ "close enough" to ρ
- "Ignore" that we replan all the time



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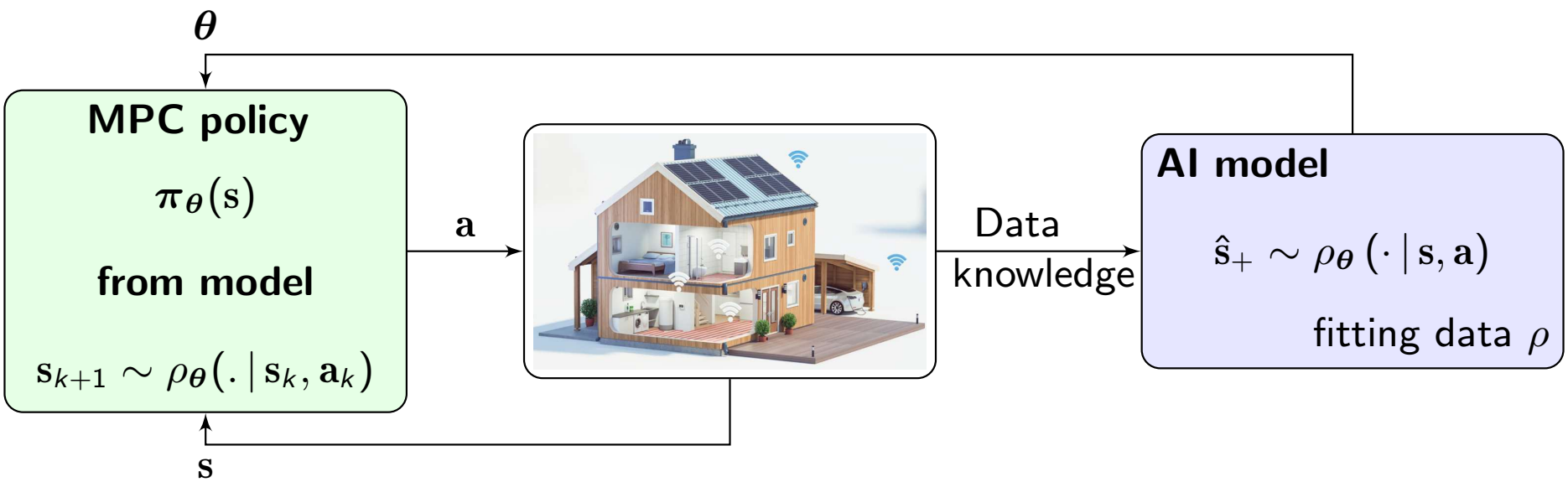
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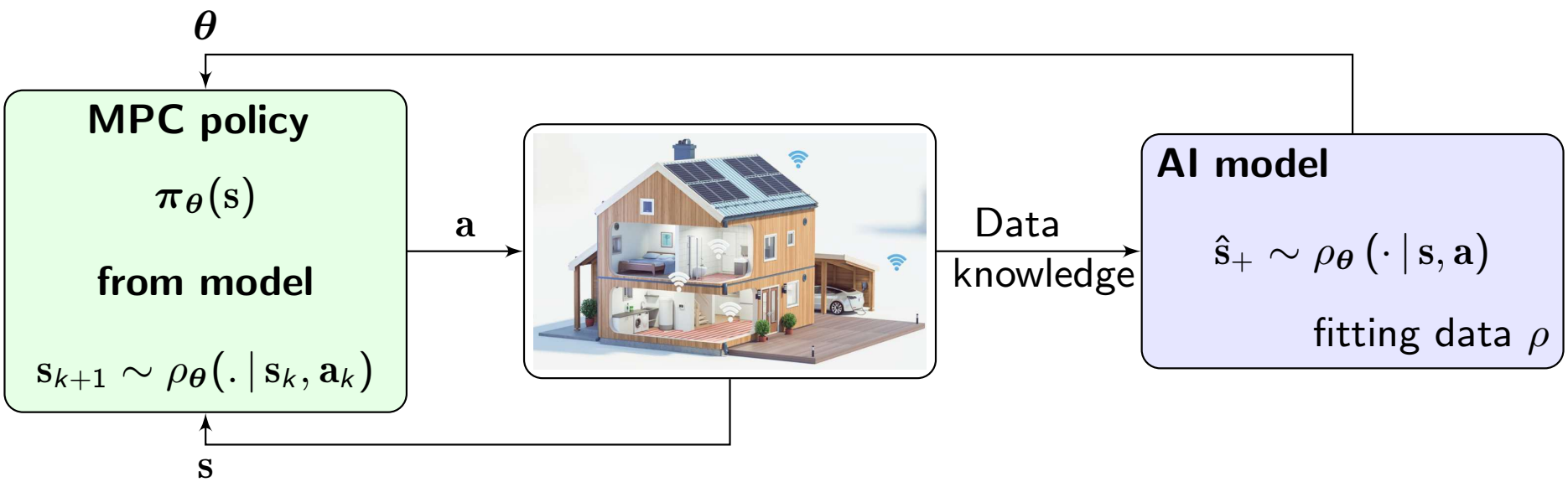
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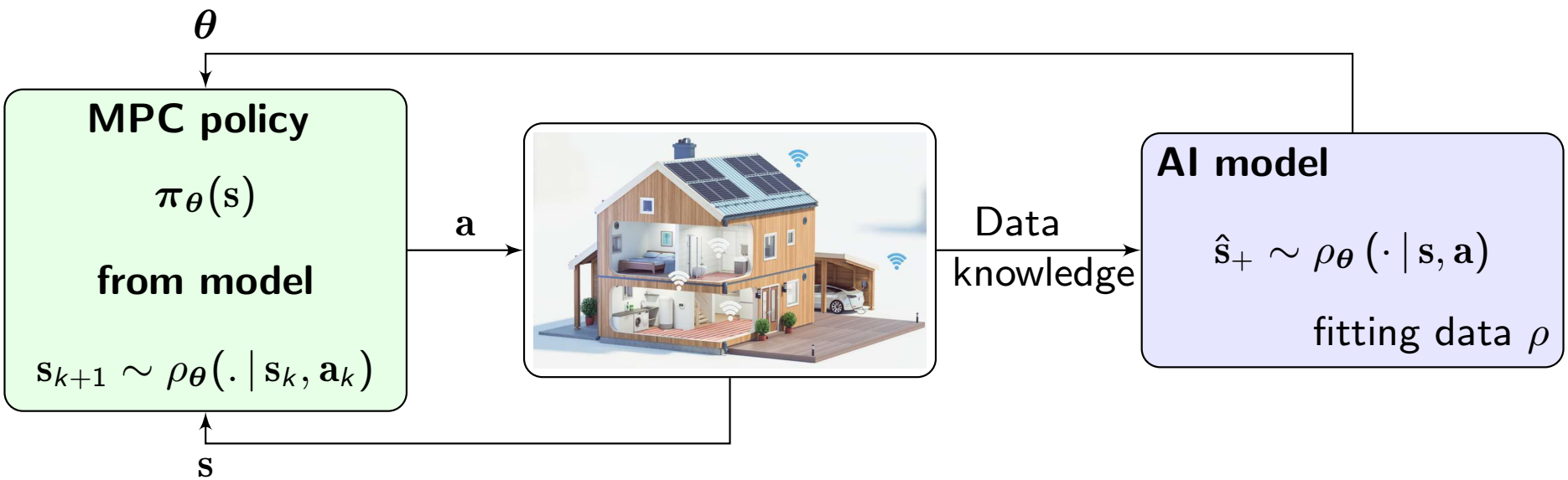
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This is not the only way of taking decisions from models

Decision policy from AI models (Sim2Real)

AI model

$$\hat{\mathbf{s}}_+ \sim \rho_{\boldsymbol{\theta}}\left(\cdot \,|\, \mathbf{s}, \mathbf{a}\right)$$

fitting data ρ

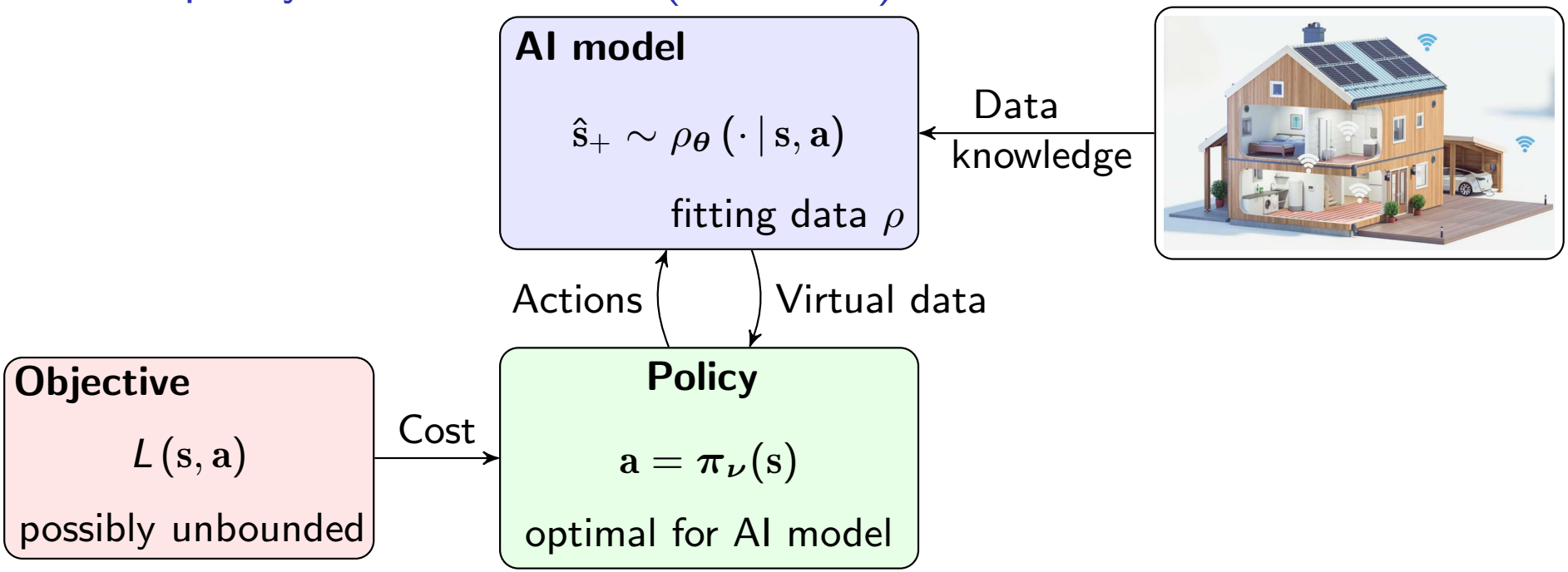
Data knowledge



Classical Process:

- Fit AI model $\rho_{\boldsymbol{\theta}}$ to real data

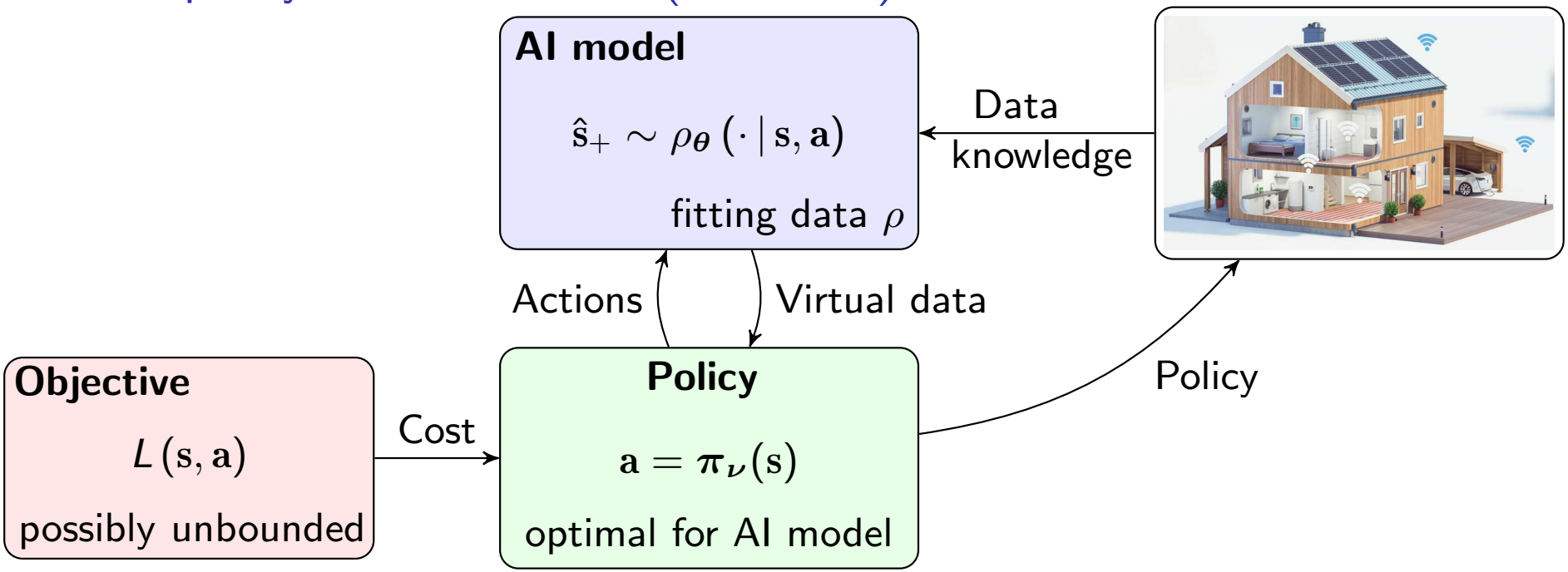
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Classical Process:

- Fit AI model $\rho_{\boldsymbol{\theta}}$ to real data
- Develop optimal policy for L and $\rho_{\boldsymbol{\theta}}$ from AI model, e.g. using In-Sim RL
  - Define parametrized policy $\boldsymbol{\pi}_{\boldsymbol{\nu}}$
  - Optimize policy parameters $\boldsymbol{\nu}$ for performance w.r.t. AI model: $\rho_{\boldsymbol{\theta}}, L \rightarrow \boldsymbol{\nu}^\star$

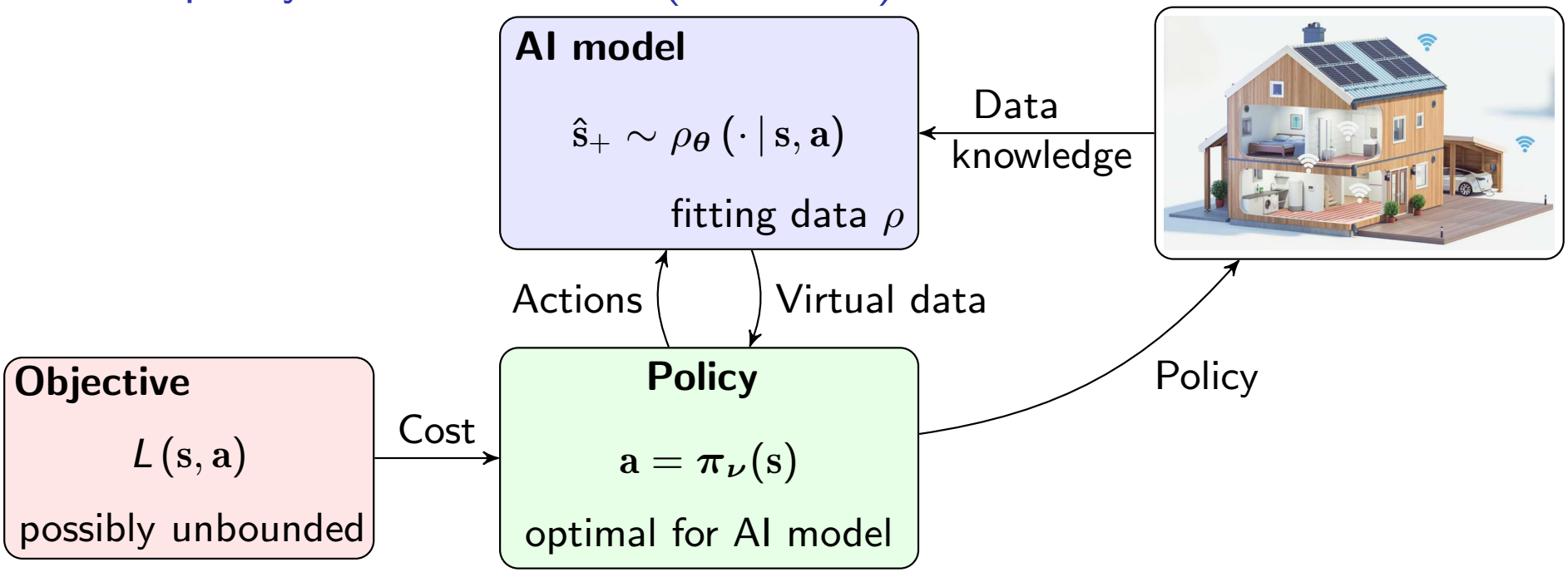
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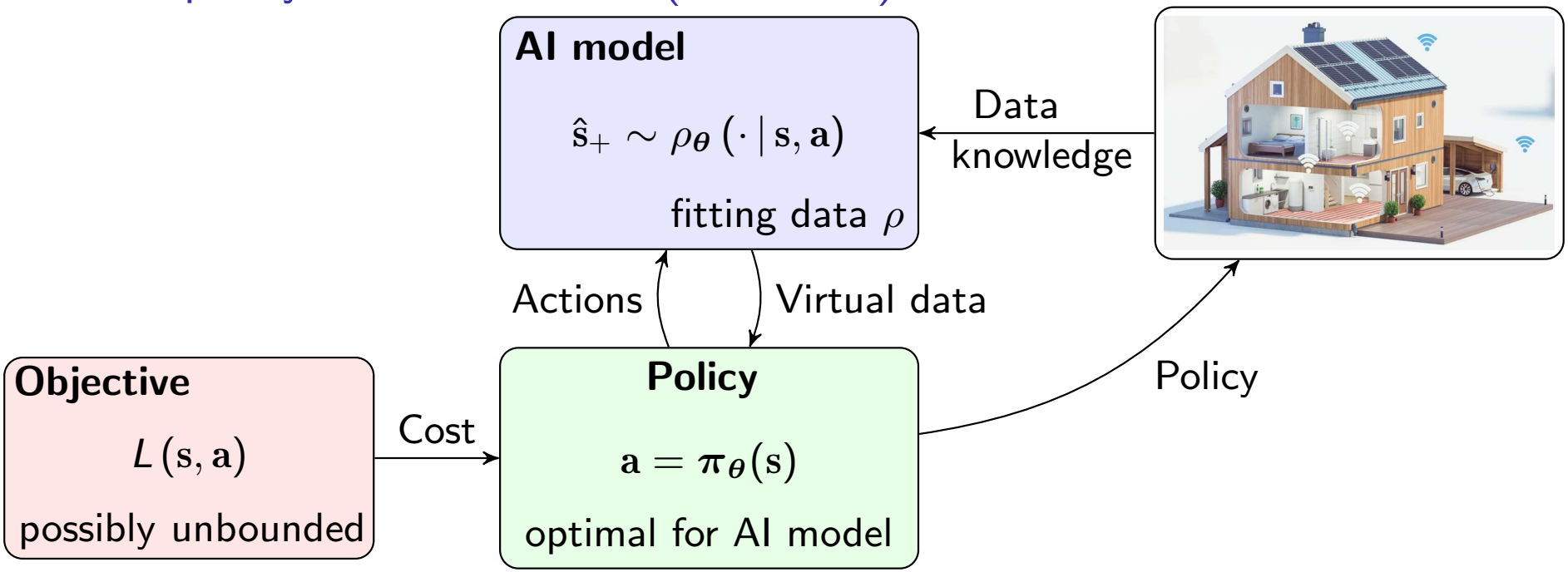


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Note: policy parameters $\nu^\star$ optimal for AI model become function of θ
Let's then label $\pi_\theta = \pi_{\nu^\star}$ for simplicity

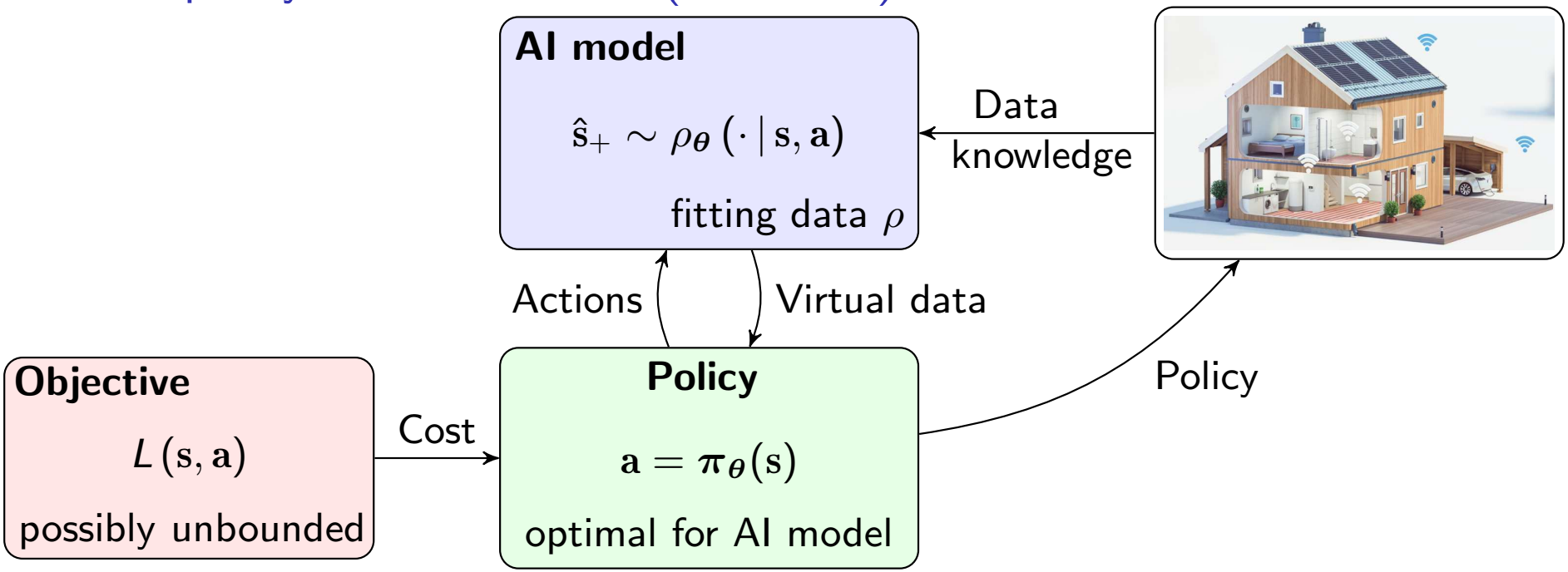
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Remarks

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  - They match if $\rho_{\boldsymbol{\theta}} = \rho$, i.e. if model is exact
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Are "standard methods" for choosing θ resulting in a good policy π_θ??
Empirically the answer is "no". Then how to choose θ to get a good policy?

MPC or Sim2Real - It's the same question...

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World MDP

World MDP definition

- States $\mathbf{s}$ and actions $\mathbf{a}$
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- Transition $\mathbf{s}_+ \sim \rho[\cdot|\mathbf{s}, \mathbf{a}]$

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Optimal policy

$$\hat{\pi}^\star(\mathbf{s}) = \arg\min_{\mathbf{a}} \hat{Q}^\star(\mathbf{s}, \mathbf{a})$$

Theory says that - under some technical conditions - there is a $\hat{L}$ such that $\hat{Q}^\star = Q^\star$

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Optimal policy

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Theory says that - under some technical conditions - there is a $\hat{L}$ such that $\hat{Q}^\star = Q^\star$

Proof: telescopic sum, some non-trivial assumptions to prevent $\infty - \infty$ cancellations

More on MDP Equivalence

World MDP

- States $\mathbf{s}$ and actions $\mathbf{a}$
- Cost $L(\mathbf{s}, \mathbf{a})$
- Transition $\mathbf{s}_+ \sim \rho[\cdot|\mathbf{s}, \mathbf{a}]$

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In-Sim policy training uses a Model MDP

- Practitioners mostly work on model $\hat{\rho}$, but "unsure" on how to tune it
- In the ML communities, people talk about "value alignment", we are trying to make this "MDP equivalence" understood

More on MDP Equivalence

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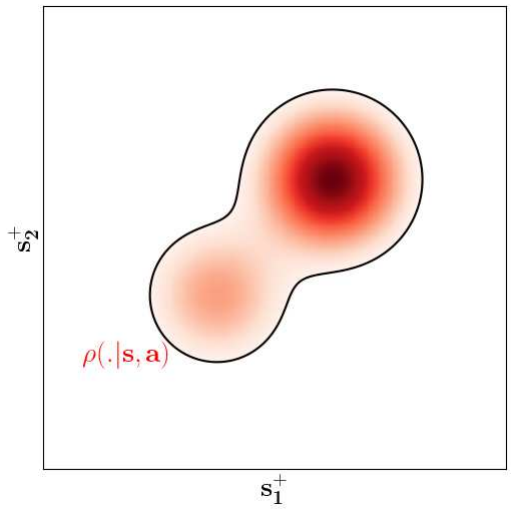
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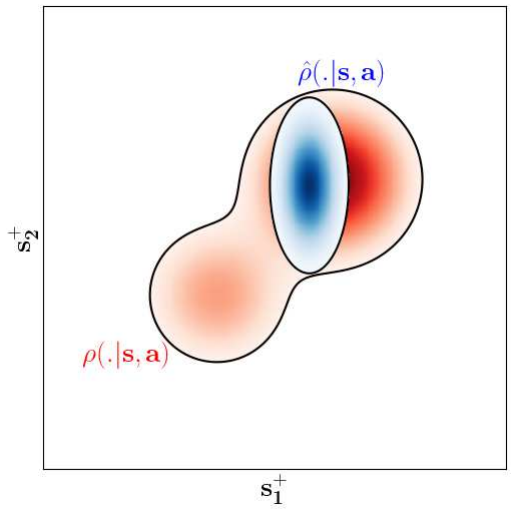
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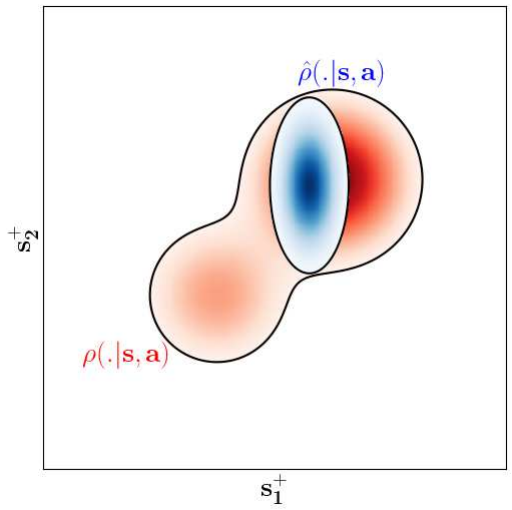
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- If $V^\star$ is continuous and support of ρ is bounded(?) and path-connected, then we can have support $\hat{\rho} \subset$ support of ρ
- *More simply said*: we can build optimal models $\hat{\rho}$ that make "plausible" predictions about the real world.



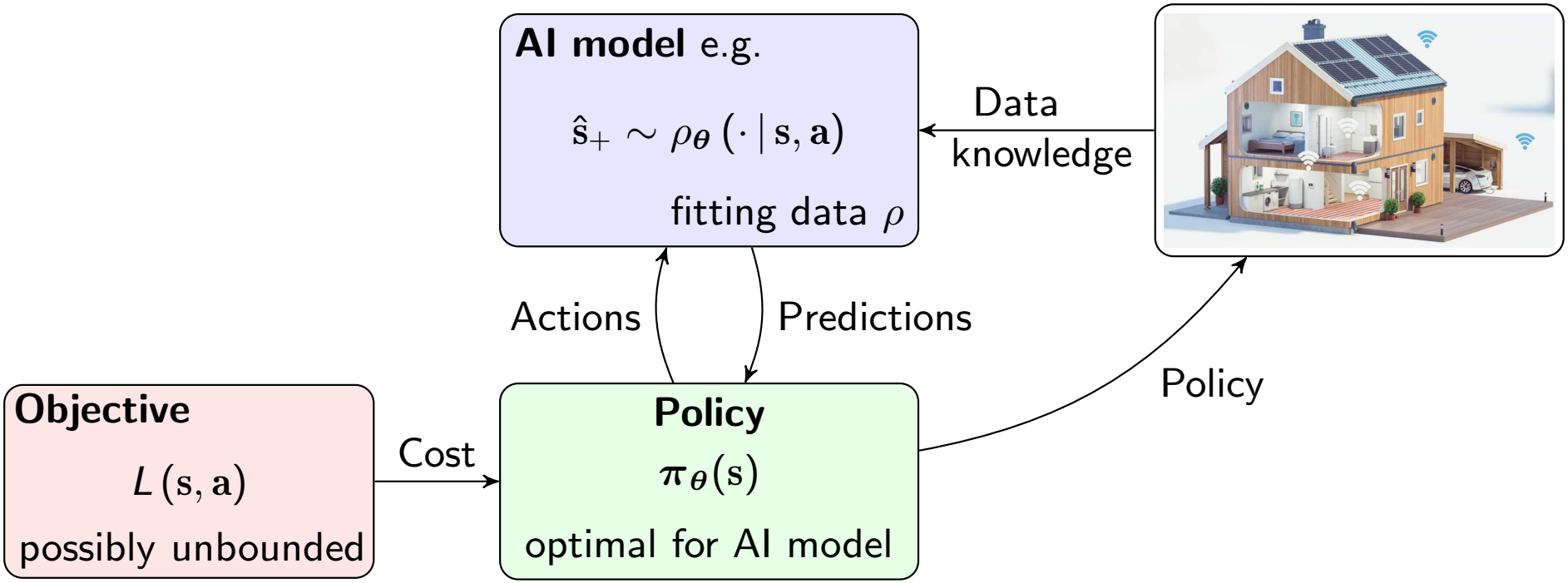
Outline

1 Decisions from data

2 AI models for Decision



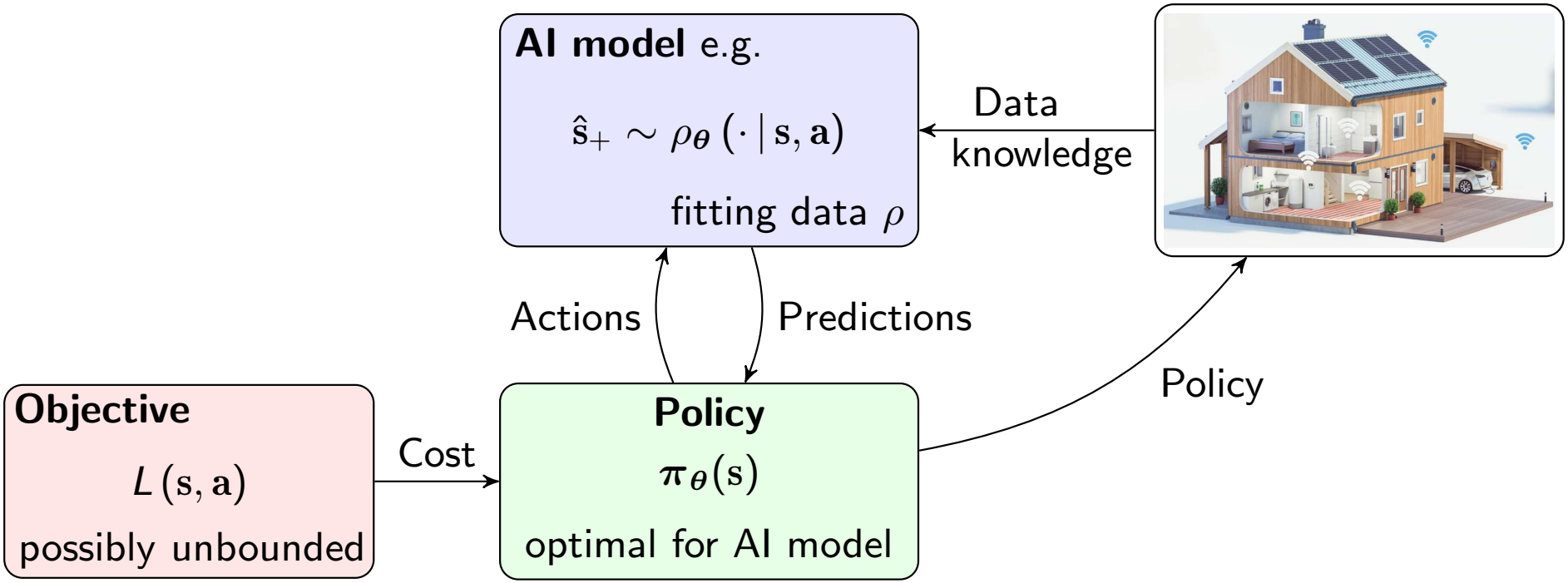
AI models for Decisions?



Learning model ρ_θ should aim at identifying

$$\boldsymbol{\theta}^\star = \arg\min_{\boldsymbol{\theta}} J(\pi_{\boldsymbol{\theta}}) \tag{1}$$

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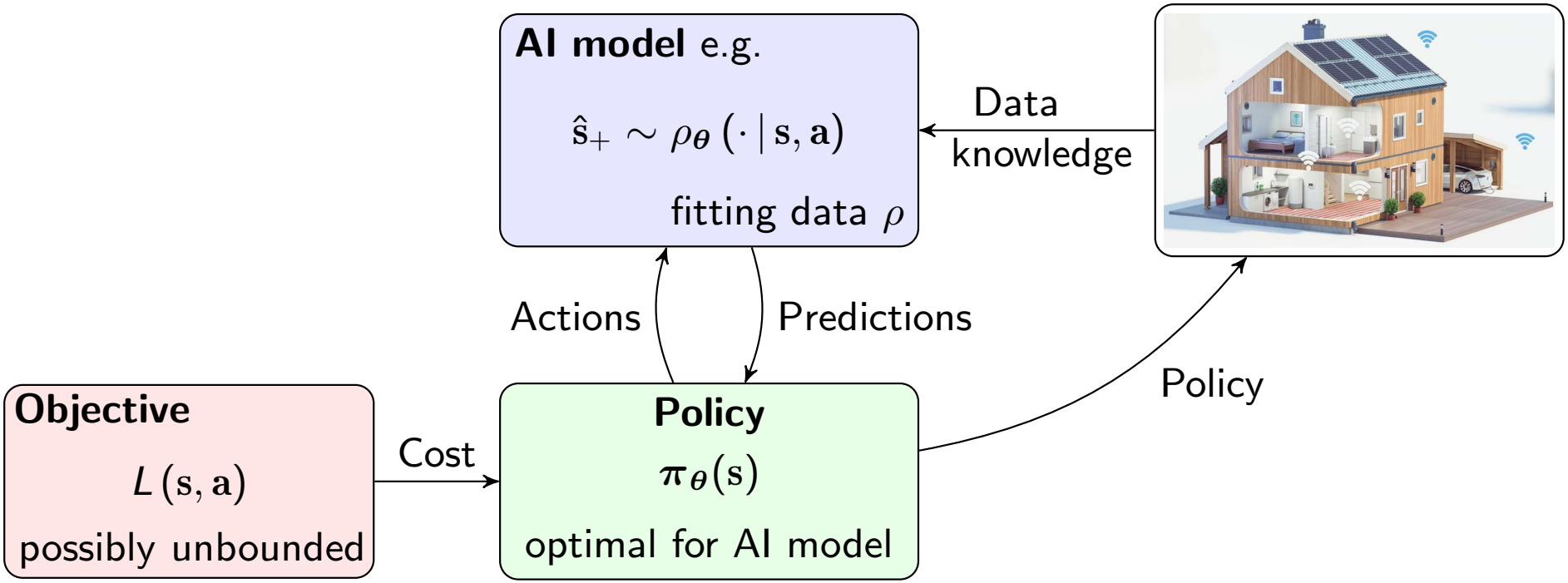


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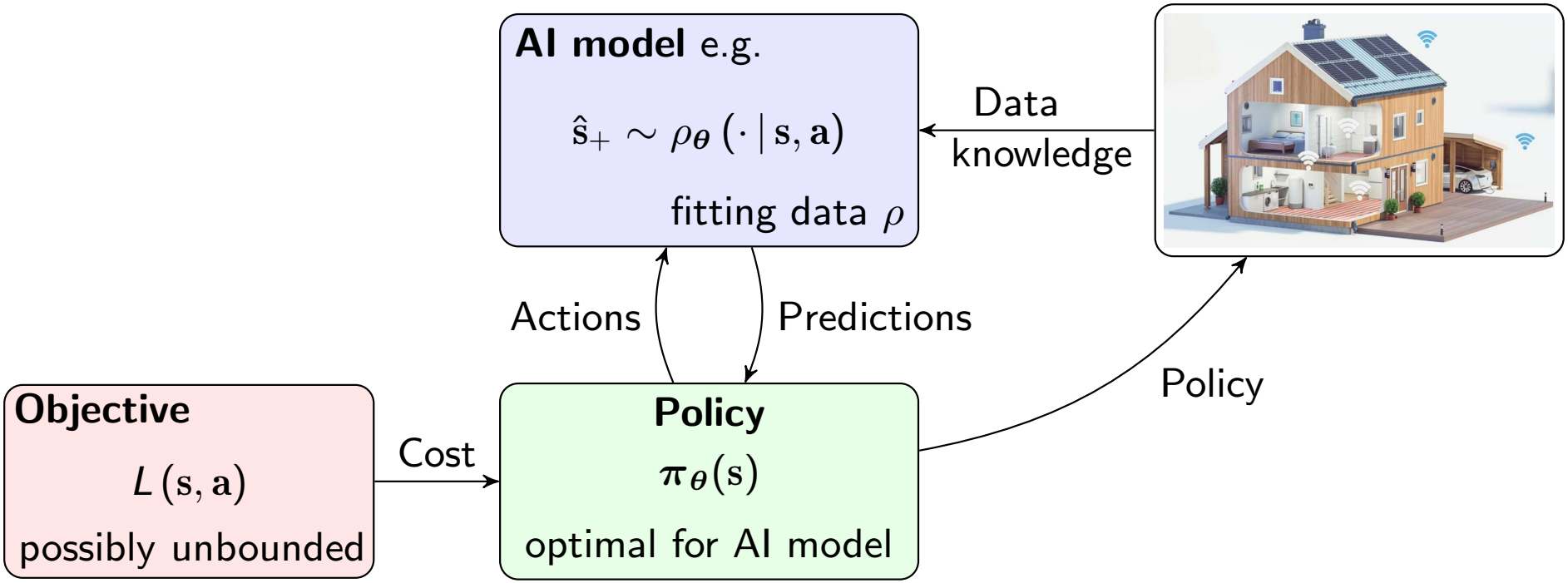


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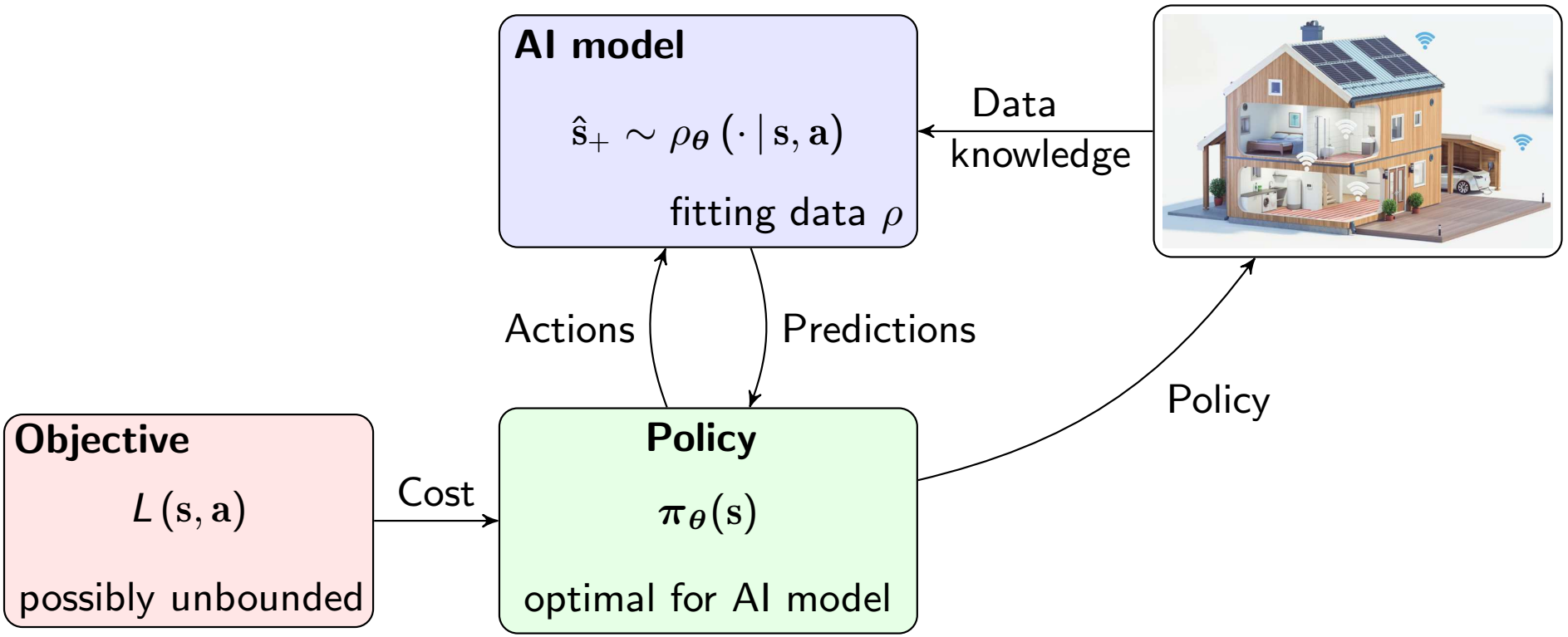


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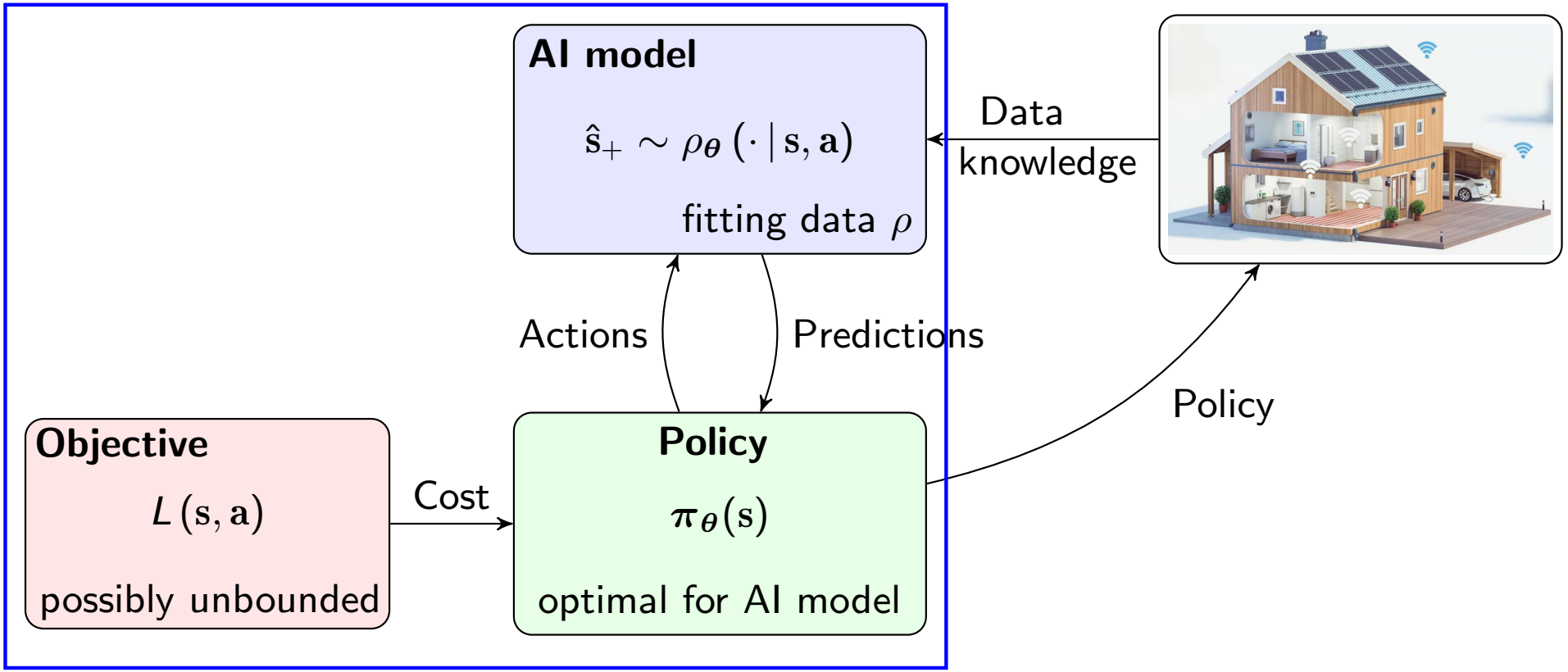
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- $\nabla_{\boldsymbol{\theta}} J(\boldsymbol{\pi}_{\boldsymbol{\theta}})$ is to be estimated from data, i.e. data hungry, noisy, possible biases
- (1) is in general different than model fitting, i.e. no loss function does (1)

Optimal AI models for Decision - A Paradigm Shift

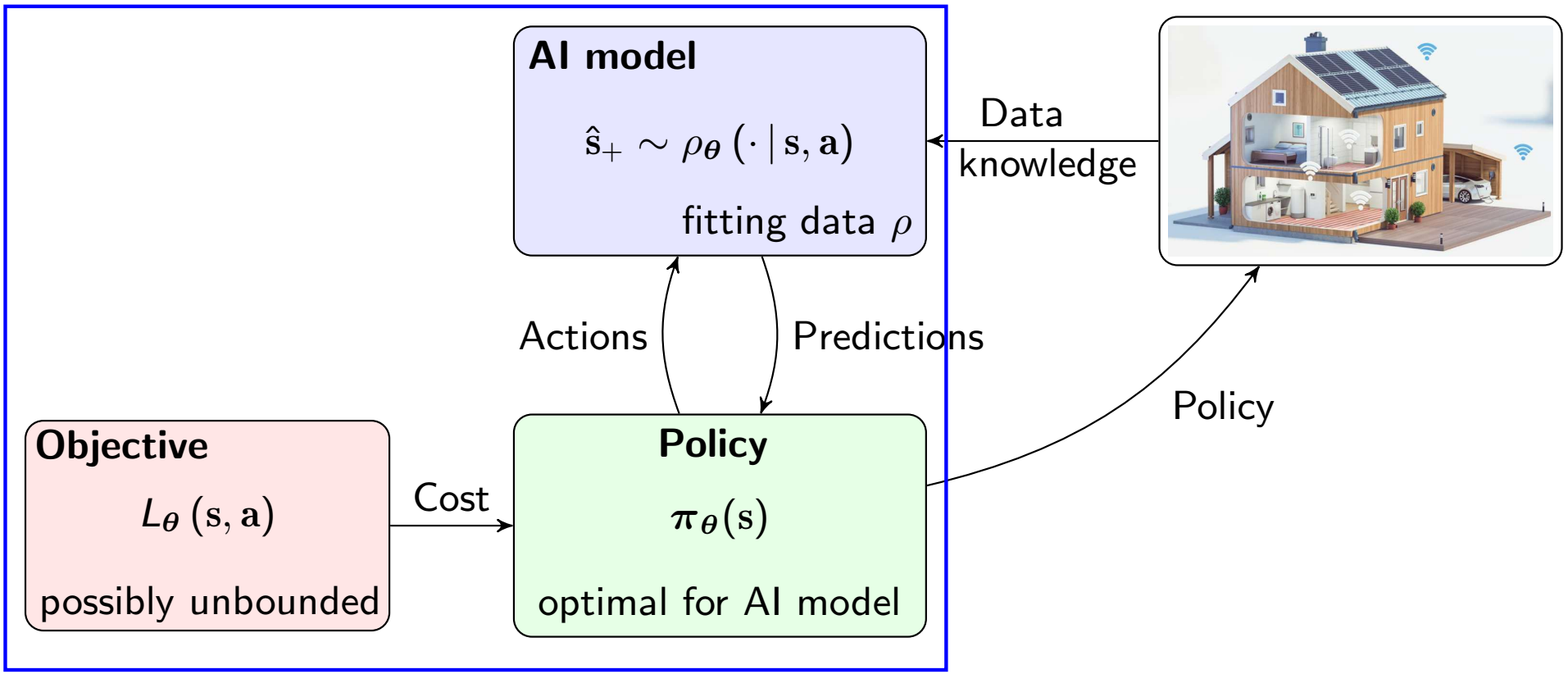


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- The model is not $\rho_{\boldsymbol{\theta}}$, it is the entire "decision-making box"

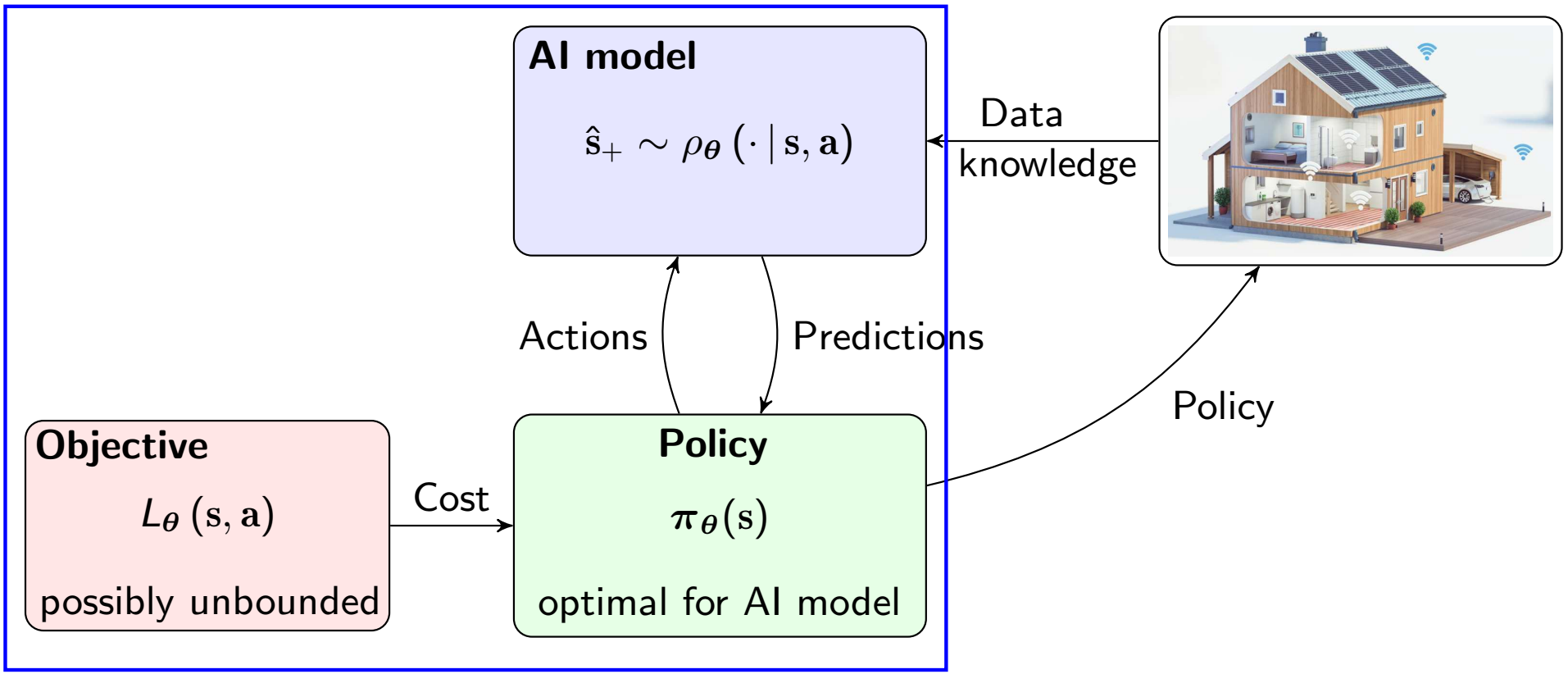
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- The model includes objective $L_{\boldsymbol{\theta}}$ used to build the policy*

* policy performance on real system still assessed via L

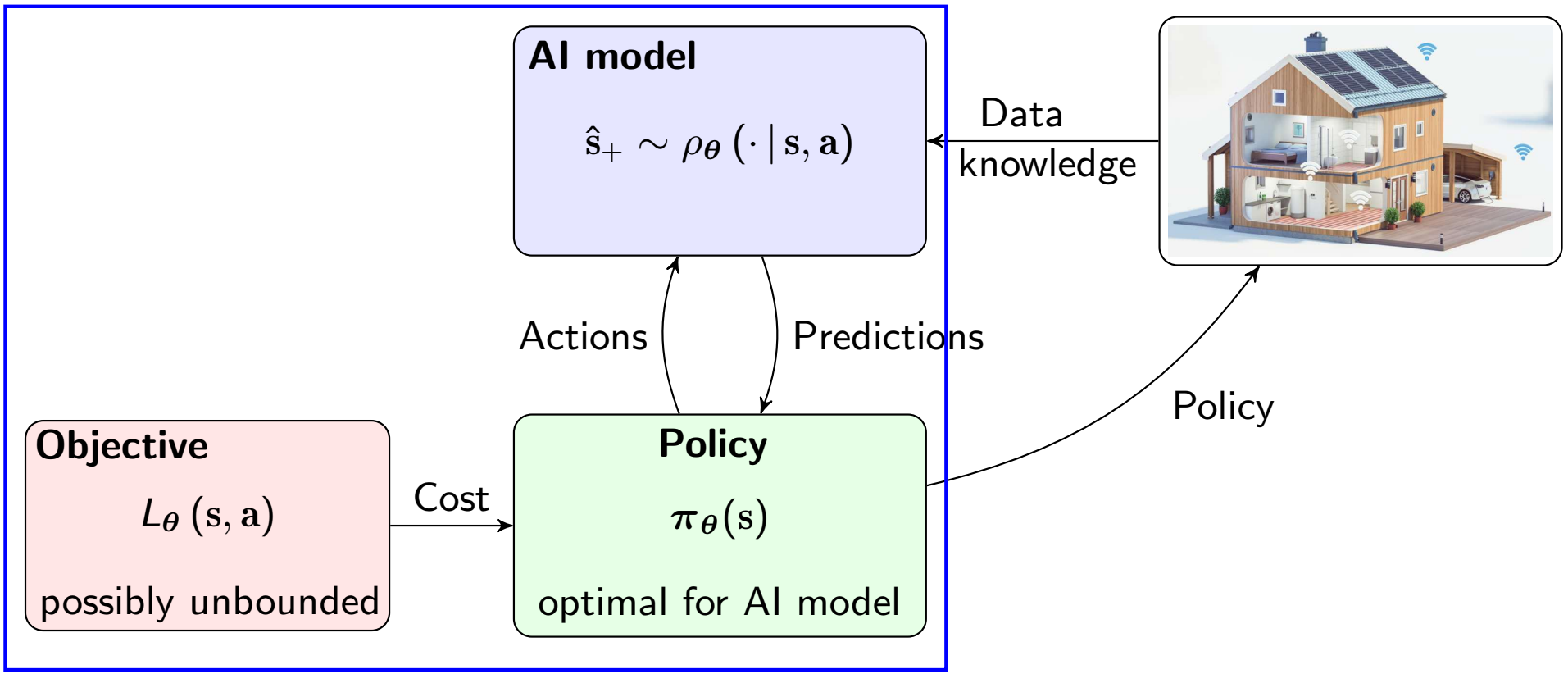
Optimal AI models for Decision - A Paradigm Shift



- The model is not $\rho_{\boldsymbol{\theta}}$, it is the entire "decision-making box"
- The model includes objective $L_{\boldsymbol{\theta}}$ used to build the policy*
- Best model $\rho_{\boldsymbol{\theta}}$ should *not necessarily* represent the data in a "classical sense"

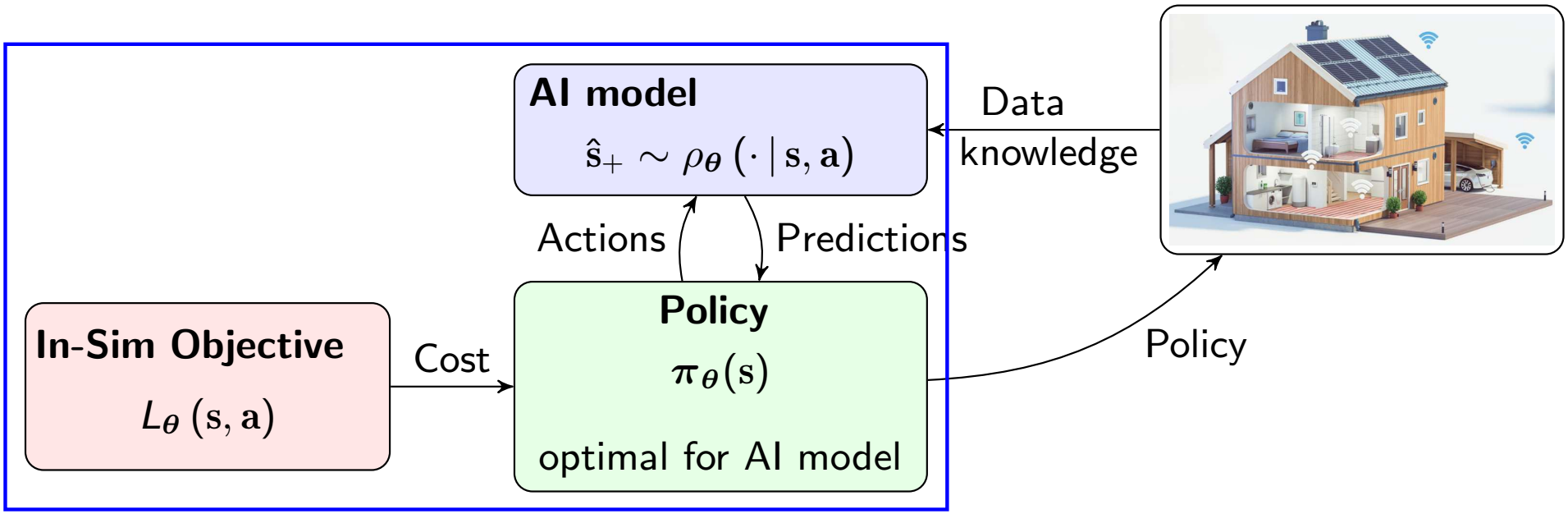
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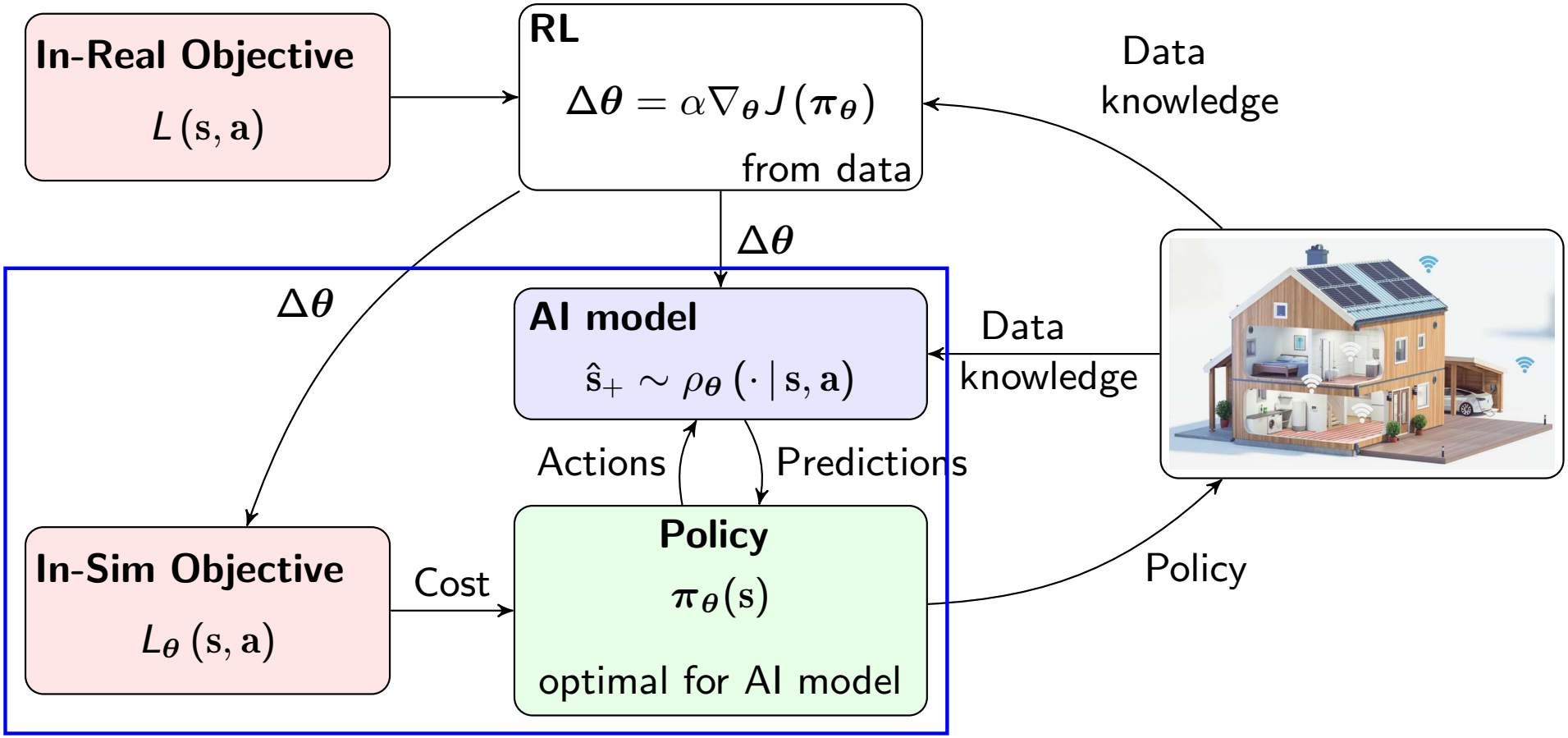


Theorem: under some (technical) assumptions, there is a $L_{\boldsymbol{\theta}}$ such that $\boldsymbol{\pi}_{\boldsymbol{\theta}} = \boldsymbol{\pi}_\star$, even if $\rho_{\boldsymbol{\theta}}$ does not represent real world ρ correctly

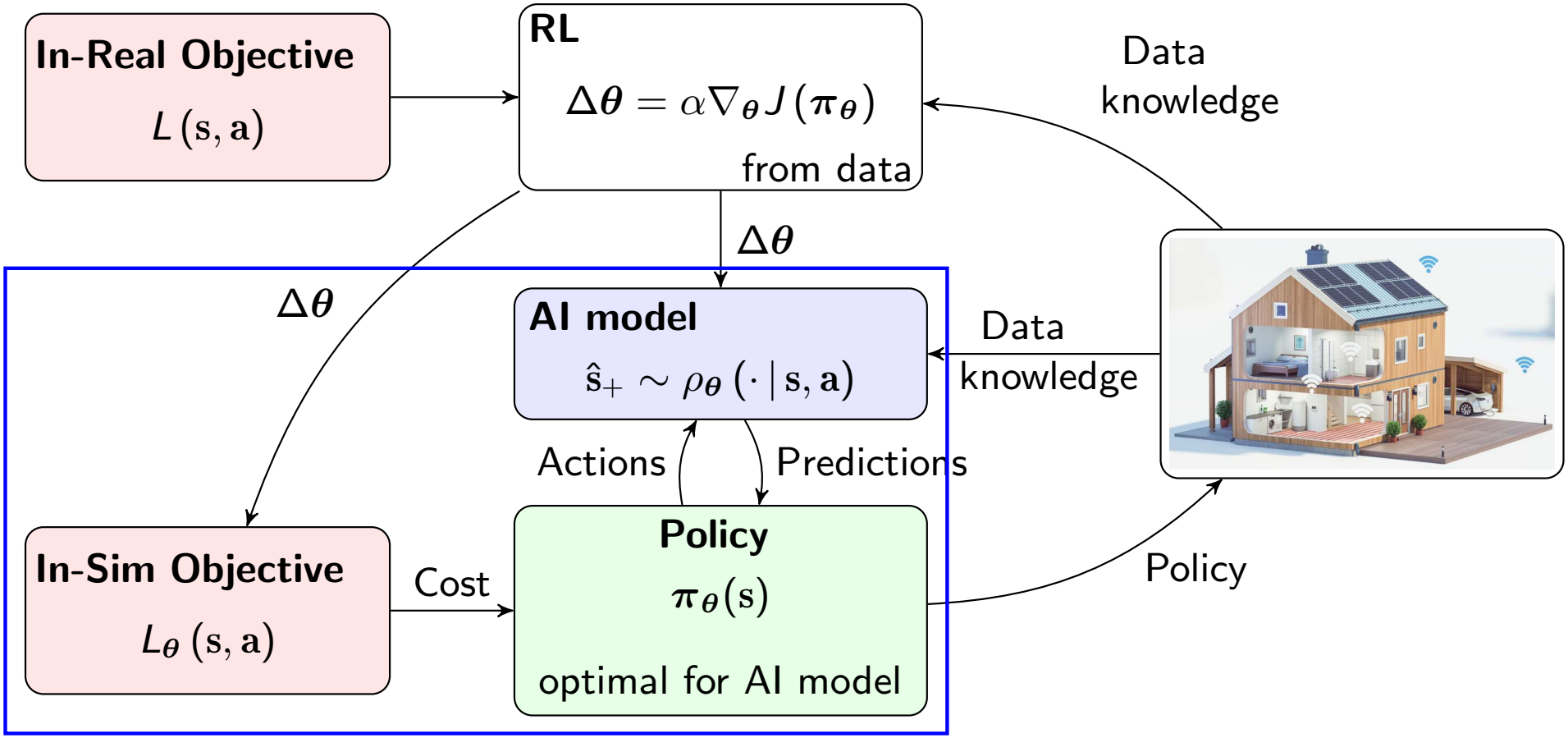
How to use this more generally? RL over Policy Training



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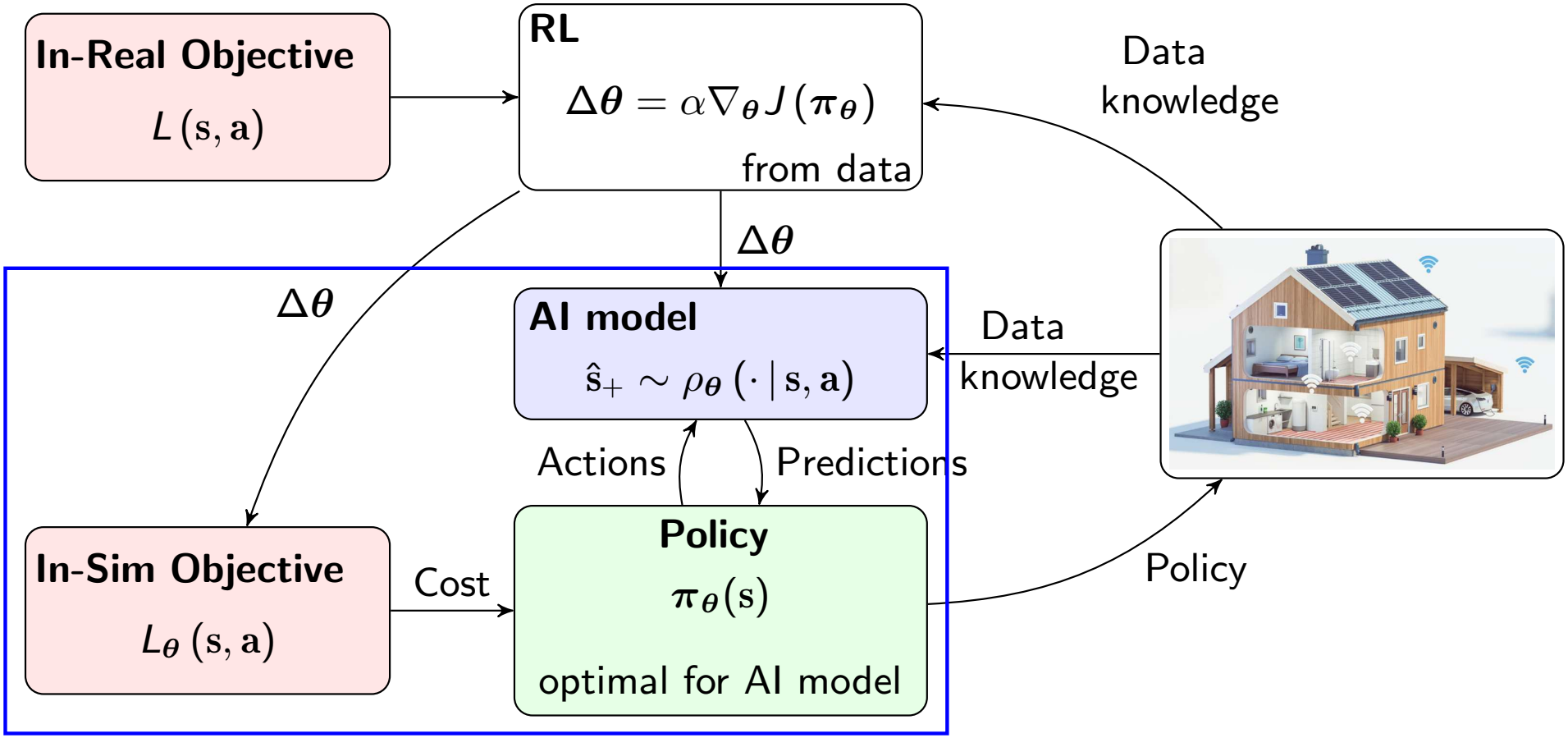


Policy gradient

$$\nabla_{\boldsymbol{\theta}} J(\boldsymbol{\pi}_{\boldsymbol{\theta}}) = \mathbb{E}\left[\nabla_{\boldsymbol{\theta}}\boldsymbol{\pi}_{\boldsymbol{\theta}} \nabla_{\mathbf{a}} Q^{\boldsymbol{\pi}_{\boldsymbol{\theta}}}\right]$$

- $Q^{\boldsymbol{\pi}_{\boldsymbol{\theta}}}$ is the critic, well-established RL tool
- $\nabla_{\boldsymbol{\theta}}\boldsymbol{\pi}_{\boldsymbol{\theta}}$ requires differentiating the closed-loop simulation...

How to use this more generally? RL over Policy Training



Difficulty: computing $\nabla_{\boldsymbol{\theta}} \boldsymbol{\pi}_{\boldsymbol{\theta}}$ requires total differentiation through the simulation-based policy optimization process:

$\boldsymbol{\theta} \rightarrow$ simulations $\rightarrow$ optimal policy for $\rho_{\boldsymbol{\theta}}$

Differentiating through policy optimization can be computationally heavy

Policy Differentiation

Performance for a stochastic policy $\pi_{\boldsymbol{\nu}}$, model MDP defined by $L_{\boldsymbol{\theta}}$, $\hat{\rho}_{\boldsymbol{\theta}}$, γ

$$\hat{J}_{\boldsymbol{\theta}}\left(\pi_{\boldsymbol{\nu}}\right) = \mathbb{E}_{\hat{\rho}_{\boldsymbol{\theta}}}\left[\sum_{k=0}^{N} \gamma^k L_{\boldsymbol{\theta}}\left(\mathbf{s}_k, \mathbf{a}_k\right) \,\middle|\, \mathbf{a}_k \sim \pi_{\boldsymbol{\nu}}\left[\cdot|\mathbf{s}_k\right]\right]$$

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Policy sensitivity: if we change $\boldsymbol{\theta}$ (cost $L_{\boldsymbol{\theta}}$ and simulation $\hat{\rho}_{\boldsymbol{\theta}}$) how does $\pi_{\boldsymbol{\nu}}$ change?

$$\frac{\partial^2}{\partial \boldsymbol{\nu}^2} \hat{J}_{\boldsymbol{\theta}}\left(\pi_{\boldsymbol{\nu}}\right) \underbrace{\frac{\mathrm{d}\pi_{\boldsymbol{\nu}}}{\mathrm{d}\boldsymbol{\theta}}}_{=\nabla_{\boldsymbol{\theta}} \pi_{\boldsymbol{\theta}}} + \frac{\partial^2}{\partial \boldsymbol{\nu} \partial \boldsymbol{\theta}} \hat{J}_{\boldsymbol{\theta}}\left(\pi_{\boldsymbol{\nu}}\right) = 0$$

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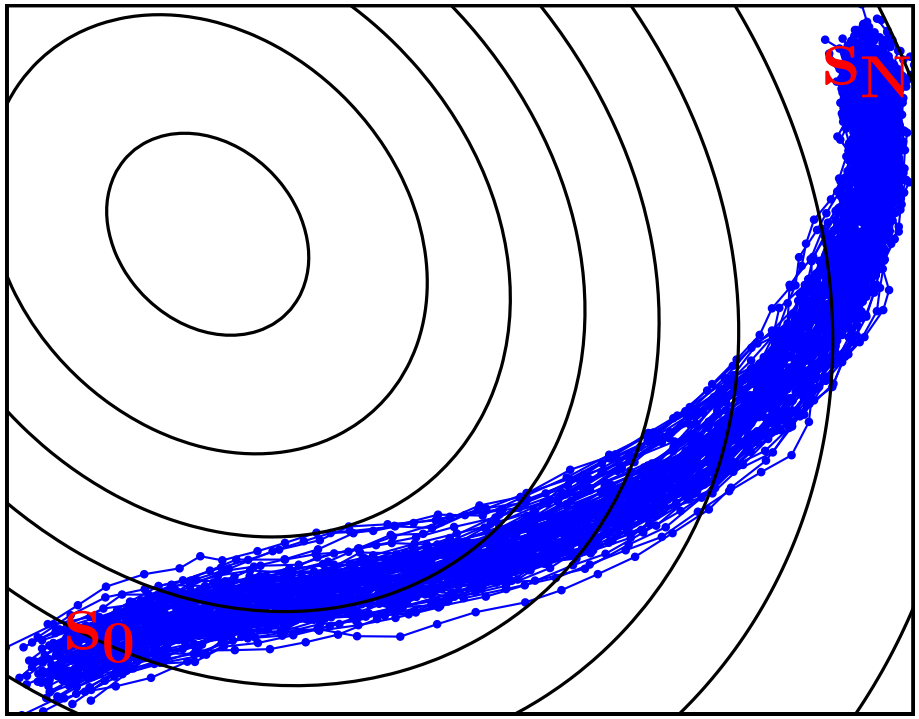
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Simulation: (with $\mathbf{s} \equiv \mathbf{s}_{0,\ldots,N}$ and $\mathbf{a} \equiv \mathbf{a}_{0,\ldots,N}$)

$$\hat{J}_{\boldsymbol{\theta}}\left(\pi_{\boldsymbol{\nu}}\right) = \int \sum_{k=0}^{N} \gamma^k L_{\boldsymbol{\theta}}\left(\mathbf{s}_k, \mathbf{a}_k\right) \varphi\left(\mathbf{s}, \mathbf{a}\right) \mathrm{d}\mathbf{s}\mathrm{d}\mathbf{a}$$

$$\varphi\left(\mathbf{s}, \mathbf{a}\right) = \rho_0\left[\mathbf{s}_0\right] \prod_{k=0}^{N-1} \hat{\rho}_{\boldsymbol{\theta}}\left[\mathbf{s}_{k+1}|\mathbf{s}_k, \mathbf{a}_k\right] \prod_{k=0}^{N} \pi_{\boldsymbol{\nu}}\left[\mathbf{a}_k|\mathbf{s}_k\right] \qquad \text{(density of the simulation)}$$

Policy Differentiation



- Blue trajectories are realization of φ:

$$\mathbf{s}^i, \mathbf{a}^i \sim \varphi(\cdot, \cdot)$$

- Sample-based evaluation (n samples)

$$\hat{J}_{\boldsymbol{\theta}}(\pi_{\boldsymbol{\nu}}) \approx \frac{1}{n} \sum_{k=i}^{n} \sum_{k=0}^{N} \gamma^k L_{\boldsymbol{\theta}}\left(\mathbf{s}_k^i, \mathbf{a}_k^i\right)$$

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$$\hat{J}_{\boldsymbol{\theta}}(\pi_{\boldsymbol{\nu}}) = \int \sum_{k=0}^{N} \gamma^k L_{\boldsymbol{\theta}}(\mathbf{s}_k, \mathbf{a}_k) \varphi(\mathbf{s}, \mathbf{a}) \, \mathrm{d}\mathbf{s}\mathrm{d}\mathbf{a}$$

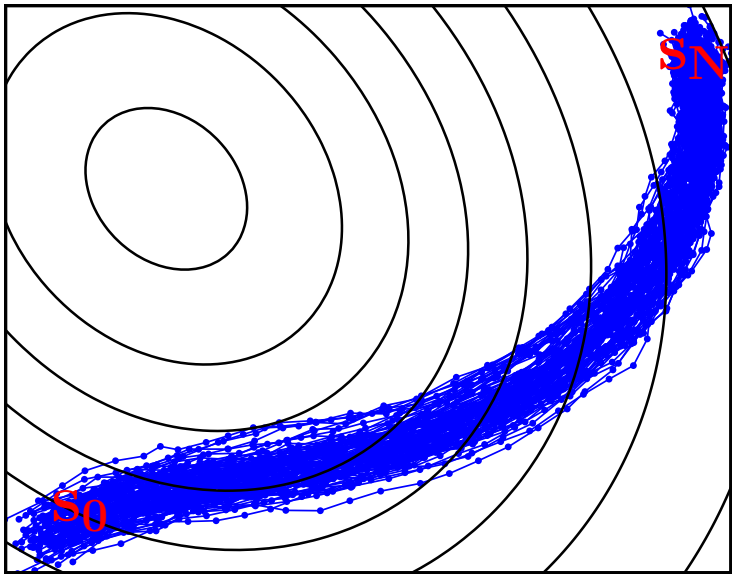
$$\varphi(\mathbf{s}, \mathbf{a}) = \rho_0[\mathbf{s}_0] \prod_{k=0}^{N-1} \hat{\rho}_{\boldsymbol{\theta}}[\mathbf{s}_{k+1} | \mathbf{s}_k, \mathbf{a}_k] \prod_{k=0}^{N} \pi_{\boldsymbol{\nu}}[\mathbf{a}_k | \mathbf{s}_k] \qquad \text{(density of the simulation)}$$

Direct Simulation Differentiation

Simulation

$$\hat{J}_{\boldsymbol{\theta}}\left(\pi_{\boldsymbol{\nu}}\right) = \int \sum_{k=0}^{N} \gamma^k L_{\boldsymbol{\theta}}\left(\mathbf{s}_k, \mathbf{a}_k\right) \varphi\left(\mathbf{s}, \mathbf{a}\right) \mathrm{dsda}$$

$$\varphi\left(\mathbf{s}, \mathbf{a}\right) = \rho_0\left[\mathbf{s}_0\right] \prod_{k=0}^{N-1} \hat{\rho}_{\boldsymbol{\theta}}\left[\mathbf{s}_{k+1} | \mathbf{s}_k, \mathbf{a}_k\right] \prod_{k=0}^{N} \pi_{\boldsymbol{\nu}}\left[\mathbf{a}_k | \mathbf{s}_k\right]$$

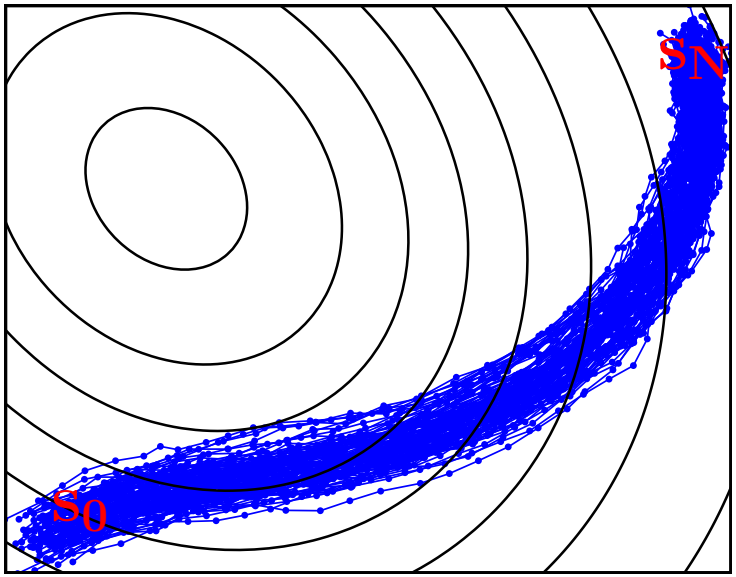


Direct Simulation Differentiation

Simulation

$$\hat{J}_{\boldsymbol{\theta}}\left(\pi_{\boldsymbol{\nu}}\right) = \int \sum_{k=0}^{N} \gamma^k L_{\boldsymbol{\theta}}\left(\mathbf{s}_k, \mathbf{a}_k\right) \varphi\left(\mathbf{s}, \mathbf{a}\right) \mathrm{dsda}$$

$$\varphi\left(\mathbf{s}, \mathbf{a}\right) = \rho_0\left[\mathbf{s}_0\right] \prod_{k=0}^{N-1} \hat{\rho}_{\boldsymbol{\theta}}\left[\mathbf{s}_{k+1} | \mathbf{s}_k, \mathbf{a}_k\right] \prod_{k=0}^{N} \pi_{\boldsymbol{\nu}}\left[\mathbf{a}_k | \mathbf{s}_k\right]$$



Direct Differentiation:

$$\partial \hat{J}_{\boldsymbol{\theta}}\left(\pi_{\boldsymbol{\nu}}\right) = \mathbb{E}_{\hat{\rho}_{\boldsymbol{\theta}}}\left[\sum_{k=0}^{N} \gamma^k \left(\partial L_{\boldsymbol{\theta}}\left(\mathbf{s}_k, \mathbf{a}_k\right) + L_{\boldsymbol{\theta}}\left(\mathbf{s}_k, \mathbf{a}_k\right) \partial \log \varphi\left(\mathbf{s}, \mathbf{a}\right) \right) \middle| \mathbf{a}_k \sim \pi_{\boldsymbol{\nu}}\left[\cdot | \mathbf{s}_k\right] \right]$$

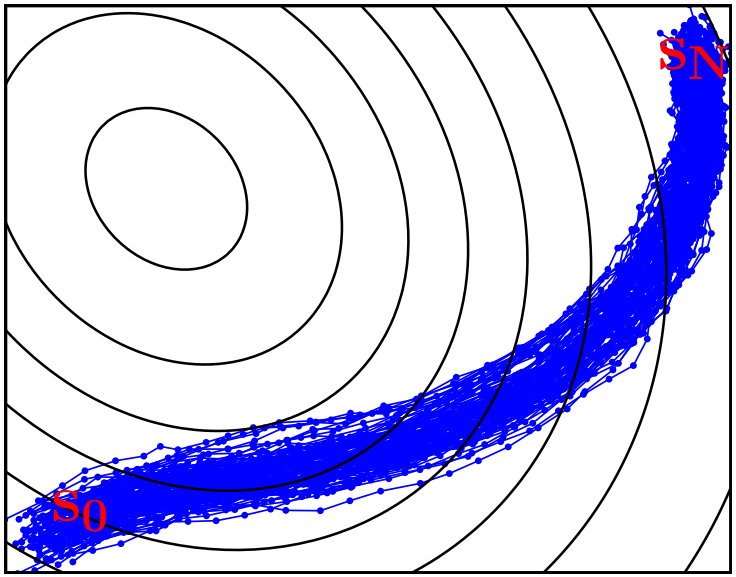
can be evaluated from data

Direct Simulation Differentiation

Simulation

$$\hat{J}_{\boldsymbol{\theta}}\left(\pi_{\boldsymbol{\nu}}\right) = \int \sum_{k=0}^{N} \gamma^k L_{\boldsymbol{\theta}}\left(\mathbf{s}_k, \mathbf{a}_k\right) \varphi\left(\mathbf{s}, \mathbf{a}\right) \mathrm{dsda}$$

$$\varphi\left(\mathbf{s}, \mathbf{a}\right) = \rho_0\left[\mathbf{s}_0\right] \prod_{k=0}^{N-1} \hat{\rho}_{\boldsymbol{\theta}}\left[\mathbf{s}_{k+1} | \mathbf{s}_k, \mathbf{a}_k\right] \prod_{k=0}^{N} \pi_{\boldsymbol{\nu}}\left[\mathbf{a}_k | \mathbf{s}_k\right]$$



Direct Differentiation:

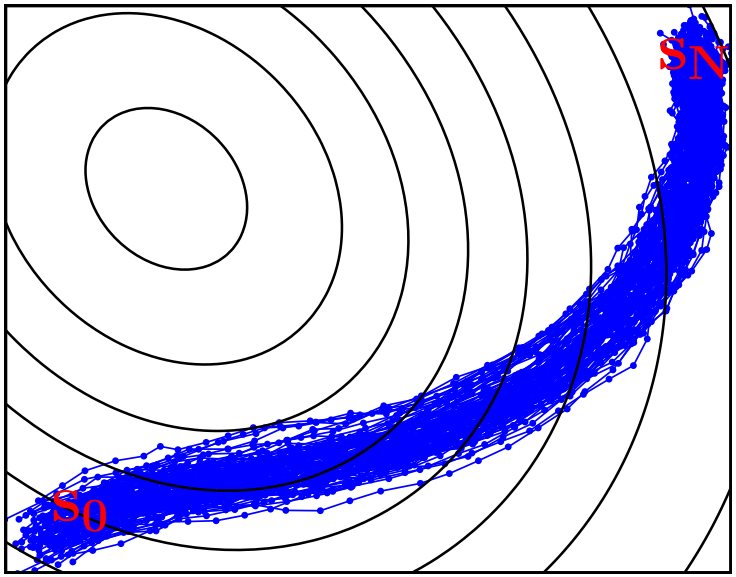
$$\partial \hat{J}_{\boldsymbol{\theta}}\left(\pi_{\boldsymbol{\nu}}\right) \approx \frac{1}{n} \sum_{i=1}^{n} \sum_{k=0}^{N} \gamma^k \left(\partial L_{\boldsymbol{\theta}}\left(\mathbf{s}_k^i, \mathbf{a}_k^i\right) + L_{\boldsymbol{\theta}}\left(\mathbf{s}_k^i, \mathbf{a}_k^i\right) \partial \log \varphi\left(\mathbf{s}_k^i, \mathbf{a}_k^i\right) \right)$$

Direct Simulation Differentiation

Simulation

$$\hat{J}_{\boldsymbol{\theta}}\left(\pi_{\boldsymbol{\nu}}\right) = \int \sum_{k=0}^{N} \gamma^k L_{\boldsymbol{\theta}}\left(\mathbf{s}_k, \mathbf{a}_k\right) \varphi\left(\mathbf{s}, \mathbf{a}\right) \mathrm{dsda}$$

$$\varphi\left(\mathbf{s}, \mathbf{a}\right) = \rho_0\left[\mathbf{s}_0\right] \prod_{k=0}^{N-1} \hat{\rho}_{\boldsymbol{\theta}}\left[\mathbf{s}_{k+1} | \mathbf{s}_k, \mathbf{a}_k\right] \prod_{k=0}^{N} \pi_{\boldsymbol{\nu}}\left[\mathbf{a}_k | \mathbf{s}_k\right]$$



Direct Differentiation:

$$\partial \hat{J}_{\boldsymbol{\theta}}\left(\pi_{\boldsymbol{\nu}}\right) \approx \frac{1}{n} \sum_{i=1}^{n} \sum_{k=0}^{N} \gamma^k \left(\partial L_{\boldsymbol{\theta}}\left(\mathbf{s}_k^i, \mathbf{a}_k^i\right) + L_{\boldsymbol{\theta}}\left(\mathbf{s}_k^i, \mathbf{a}_k^i\right) \partial \log \varphi\left(\mathbf{s}_k^i, \mathbf{a}_k^i\right) \right)$$

Remarks

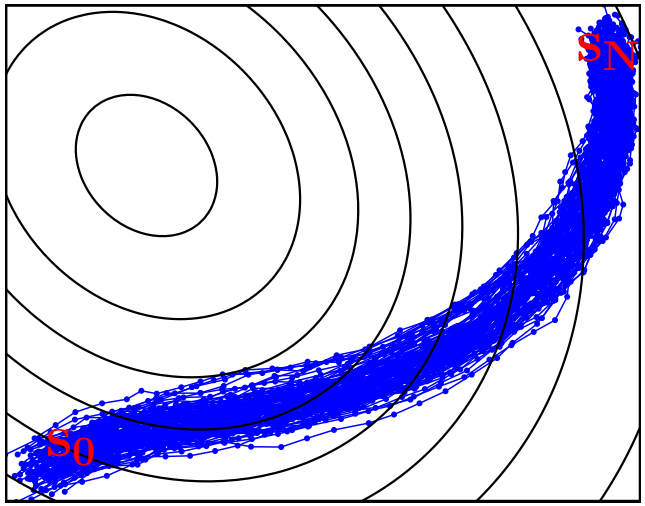
- ✓ Second-order sensitivities are similar
- ✓ Simple and computationally very efficient
- ✓ Can differentiate through discrete state and action sets
- ✘ Sample-based estimations are very noisy

Critic-Based Differentiation

In-Sim Actor-critic does

$$\nabla_{\boldsymbol{\nu}} \hat{J}_{\boldsymbol{\theta}}\left(\pi_{\boldsymbol{\nu}}\right) = \mathbb{E}_{\substack{\mathbf{s} \sim \rho_{\boldsymbol{\theta}} \\ \mathbf{a} \sim \pi_{\boldsymbol{\nu}}}} \left[\nabla_{\boldsymbol{\nu}} \log \pi_{\boldsymbol{\nu}}\left(\mathbf{a}|\mathbf{s}\right) A^{\boldsymbol{\pi}_{\boldsymbol{\nu}}}\left(\mathbf{s}, \mathbf{a}\right)\right] = 0$$

defines relationship $\pi_{\boldsymbol{\nu}^\star}$ with $\boldsymbol{\nu}^\star$ function of $\boldsymbol{\theta}$



Critic-Based Differentiation

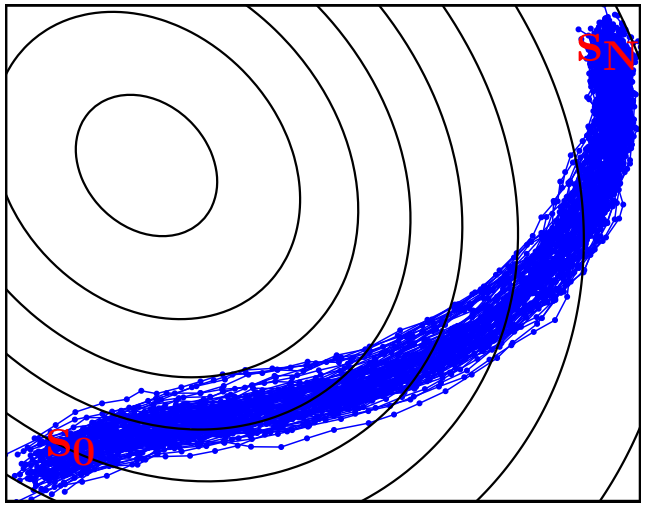
In-Sim Actor-critic does

$$\nabla_{\boldsymbol{\nu}} \hat{J}_{\boldsymbol{\theta}}\left(\pi_{\boldsymbol{\nu}}\right) = \mathbb{E}_{\substack{\mathbf{s} \sim \rho_{\boldsymbol{\theta}} \\ \mathbf{a} \sim \pi_{\boldsymbol{\nu}}}} \left[\nabla_{\boldsymbol{\nu}} \log \pi_{\boldsymbol{\nu}}\left(\mathbf{a}|\mathbf{s}\right) A^{\pi_{\boldsymbol{\nu}}}\left(\mathbf{s}, \mathbf{a}\right)\right] = 0$$

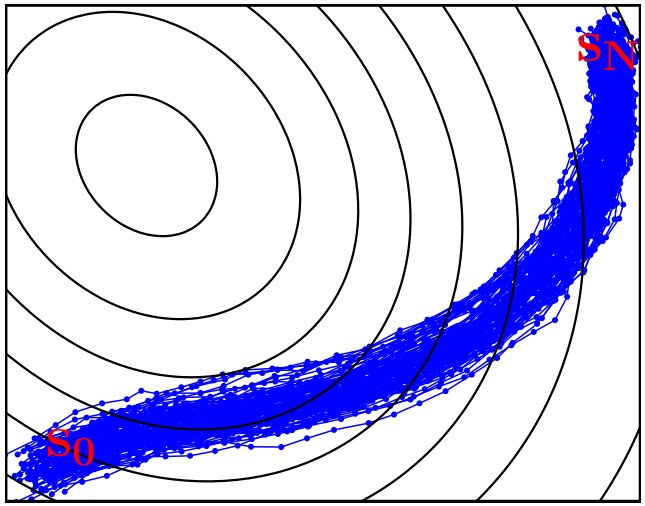
defines relationship $\pi_{\boldsymbol{\nu}^\star}$ with $\boldsymbol{\nu}^\star$ function of $\boldsymbol{\theta}$

In-Real Actor-critic evaluates

$$\nabla_{\boldsymbol{\theta}} J\left(\pi_{\boldsymbol{\nu}^\star}\right) = \mathbb{E}_{\text{Real World}}\left[\nabla_{\boldsymbol{\theta}} \log \pi_{\boldsymbol{\nu}^\star}\left(\mathbf{a}|\mathbf{s}\right) A^{\boldsymbol{\nu}^\star}\left(\mathbf{s}, \mathbf{a}\right)\right]$$



Critic-Based Differentiation



In-Sim Actor-critic does

$$\nabla_{\boldsymbol{\nu}} \hat{J}_{\boldsymbol{\theta}}\left(\pi_{\boldsymbol{\nu}}\right) = \mathbb{E}_{\substack{\mathbf{s} \sim \rho_{\boldsymbol{\theta}} \\ \mathbf{a} \sim \pi_{\boldsymbol{\nu}}}} \left[\nabla_{\boldsymbol{\nu}} \log \pi_{\boldsymbol{\nu}}\left(\mathbf{a}|\mathbf{s}\right) A^{\pi_{\boldsymbol{\nu}}}\left(\mathbf{s}, \mathbf{a}\right)\right] = 0$$

defines relationship $\pi_{\boldsymbol{\nu}^\star}$ with $\boldsymbol{\nu}^\star$ function of $\boldsymbol{\theta}$

In-Real Actor-critic evaluates

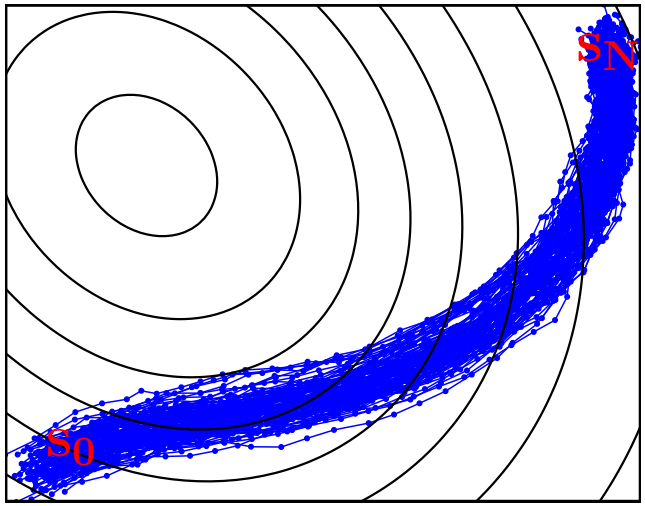
$$\nabla_{\boldsymbol{\theta}} J\left(\pi_{\boldsymbol{\nu}^\star}\right) = \mathbb{E}_{\text{Real World}}\left[\nabla_{\boldsymbol{\theta}} \log \pi_{\boldsymbol{\nu}^\star}\left(\mathbf{a}|\mathbf{s}\right) A^{\boldsymbol{\nu}^\star}\left(\mathbf{s}, \mathbf{a}\right)\right]$$

where $\nabla_{\boldsymbol{\theta}} \log \pi_{\boldsymbol{\nu}^\star}\left(\mathbf{a}|\mathbf{s}\right)$ requires $\frac{\mathrm{d}\boldsymbol{\nu}^\star}{\mathrm{d}\boldsymbol{\theta}}$

Sensitivity

$$\frac{\partial^2}{\partial \boldsymbol{\nu}^2} \hat{J}_{\boldsymbol{\theta}}\left(\pi_{\boldsymbol{\nu}^\star}\right) \frac{\mathrm{d}\boldsymbol{\nu}^\star}{\mathrm{d}\boldsymbol{\theta}} + \frac{\partial^2}{\partial \boldsymbol{\nu} \partial \boldsymbol{\theta}} \hat{J}_{\boldsymbol{\theta}}\left(\pi_{\boldsymbol{\nu}^\star}\right) = 0$$

Critic-Based Differentiation



In-Sim Actor-critic does

$$\nabla_{\boldsymbol{\nu}} \hat{J}_{\boldsymbol{\theta}}\left(\pi_{\boldsymbol{\nu}}\right) = \mathbb{E}_{\substack{\mathbf{s} \sim \rho_{\boldsymbol{\theta}} \\ \mathbf{a} \sim \pi_{\boldsymbol{\nu}}}} \left[\nabla_{\boldsymbol{\nu}} \log \pi_{\boldsymbol{\nu}}\left(\mathbf{a}|\mathbf{s}\right) A^{\pi_{\boldsymbol{\nu}}}\left(\mathbf{s}, \mathbf{a}\right)\right] = 0$$

defines relationship $\pi_{\boldsymbol{\nu}^\star}$ with $\boldsymbol{\nu}^\star$ function of $\boldsymbol{\theta}$

In-Real Actor-critic evaluates

$$\nabla_{\boldsymbol{\theta}} J\left(\pi_{\boldsymbol{\nu}^\star}\right) = \mathbb{E}_{\text{Real World}}\left[\nabla_{\boldsymbol{\theta}} \log \pi_{\boldsymbol{\nu}^\star}\left(\mathbf{a}|\mathbf{s}\right) A^{\boldsymbol{\nu}^\star}\left(\mathbf{s}, \mathbf{a}\right)\right]$$

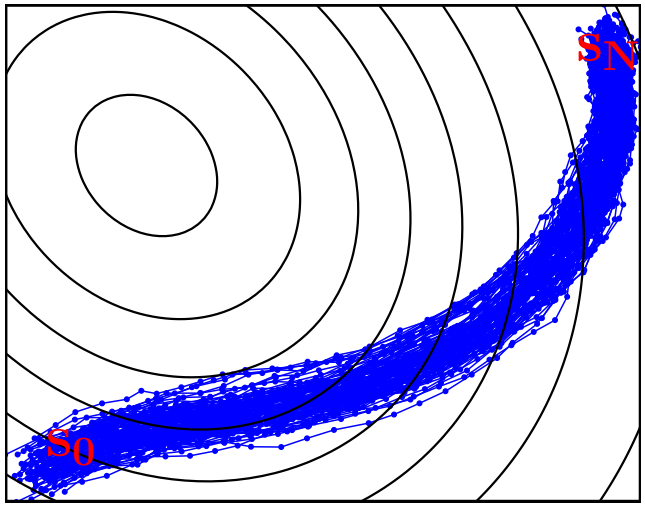
where $\nabla_{\boldsymbol{\theta}} \log \pi_{\boldsymbol{\nu}^\star}\left(\mathbf{a}|\mathbf{s}\right)$ requires $\frac{\mathrm{d}\boldsymbol{\nu}^\star}{\mathrm{d}\boldsymbol{\theta}}$

Sensitivity

$$\frac{\partial^2}{\partial \boldsymbol{\nu}^2} \hat{J}_{\boldsymbol{\theta}}\left(\pi_{\boldsymbol{\nu}^\star}\right) \frac{\mathrm{d}\boldsymbol{\nu}^\star}{\mathrm{d}\boldsymbol{\theta}} + \frac{\partial^2}{\partial \boldsymbol{\nu} \partial \boldsymbol{\theta}} \hat{J}_{\boldsymbol{\theta}}\left(\pi_{\boldsymbol{\nu}^\star}\right) = 0$$

Idea: compute $\frac{\mathrm{d}\boldsymbol{\nu}^\star}{\mathrm{d}\boldsymbol{\theta}}$ from differentiating the in-sim actor-critic w.r.t. $\frac{\partial}{\partial \boldsymbol{\nu}}$ and $\frac{\partial}{\partial \boldsymbol{\theta}}$

Critic-Based Differentiation



In-Sim Actor-critic does

$$\nabla_{\boldsymbol{\nu}} \hat{J}_{\boldsymbol{\theta}}\left(\pi_{\boldsymbol{\nu}}\right) = \mathbb{E}_{\substack{\mathbf{s} \sim \rho_{\boldsymbol{\theta}} \\ \mathbf{a} \sim \pi_{\boldsymbol{\nu}}}} \left[\nabla_{\boldsymbol{\nu}} \log \pi_{\boldsymbol{\nu}}\left(\mathbf{a}|\mathbf{s}\right) A^{\pi_{\boldsymbol{\nu}}}\left(\mathbf{s}, \mathbf{a}\right)\right] = 0$$

defines relationship $\pi_{\boldsymbol{\nu}^\star}$ with $\boldsymbol{\nu}^\star$ function of $\boldsymbol{\theta}$

In-Real Actor-critic evaluates

$$\nabla_{\boldsymbol{\theta}} J\left(\pi_{\boldsymbol{\nu}^\star}\right) = \mathbb{E}_{\text{Real World}}\left[\nabla_{\boldsymbol{\theta}} \log \pi_{\boldsymbol{\nu}^\star}\left(\mathbf{a}|\mathbf{s}\right) A^{\boldsymbol{\nu}^\star}\left(\mathbf{s}, \mathbf{a}\right)\right]$$

where $\nabla_{\boldsymbol{\theta}} \log \pi_{\boldsymbol{\nu}^\star}\left(\mathbf{a}|\mathbf{s}\right)$ requires $\frac{\mathrm{d}\boldsymbol{\nu}^\star}{\mathrm{d}\boldsymbol{\theta}}$

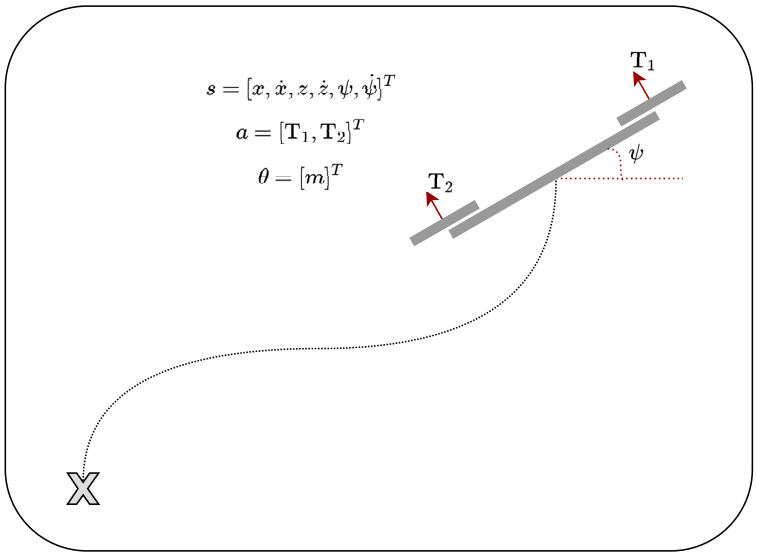
Sensitivity

$$\frac{\partial^2}{\partial \boldsymbol{\nu}^2} \hat{J}_{\boldsymbol{\theta}}\left(\pi_{\boldsymbol{\nu}^\star}\right) \frac{\mathrm{d}\boldsymbol{\nu}^\star}{\mathrm{d}\boldsymbol{\theta}} + \frac{\partial^2}{\partial \boldsymbol{\nu} \partial \boldsymbol{\theta}} \hat{J}_{\boldsymbol{\theta}}\left(\pi_{\boldsymbol{\nu}^\star}\right) = 0$$

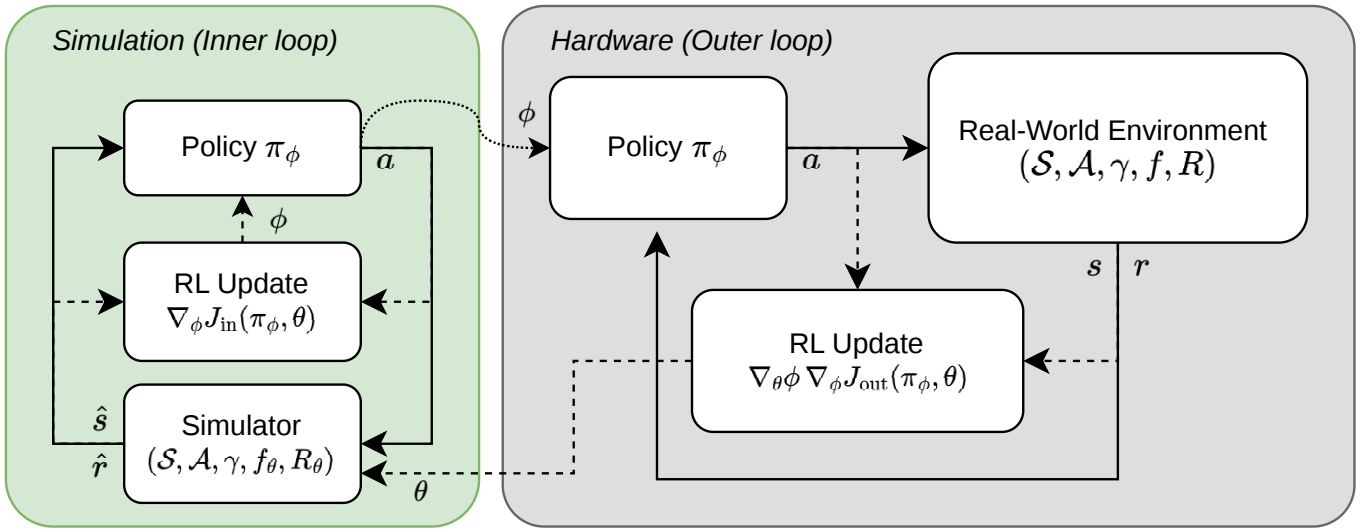
Idea: compute $\frac{\mathrm{d}\boldsymbol{\nu}^\star}{\mathrm{d}\boldsymbol{\theta}}$ from differentiating the in-sim actor-critic w.r.t. $\frac{\partial}{\partial \boldsymbol{\nu}}$ and $\frac{\partial}{\partial \boldsymbol{\theta}}$

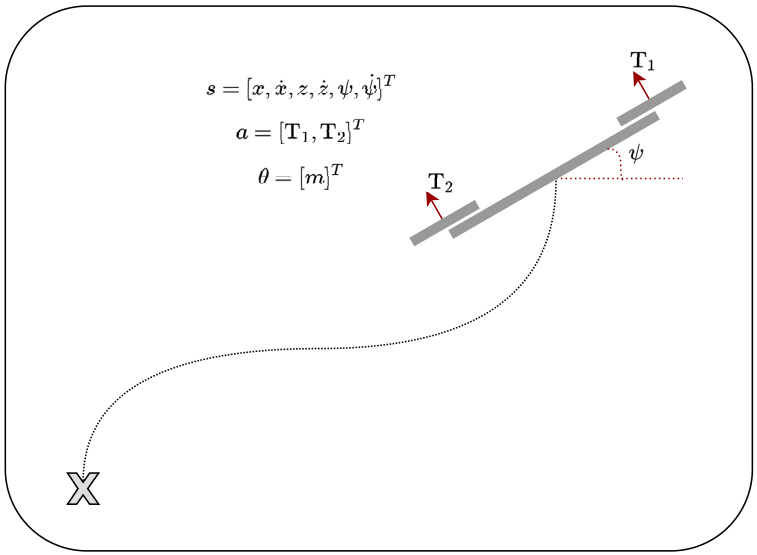
Proposed first algorithm to do that.
Computationally heavy, but we have "missed" some simplifications, testing further

Example - In-Real RL over In-Sim RL

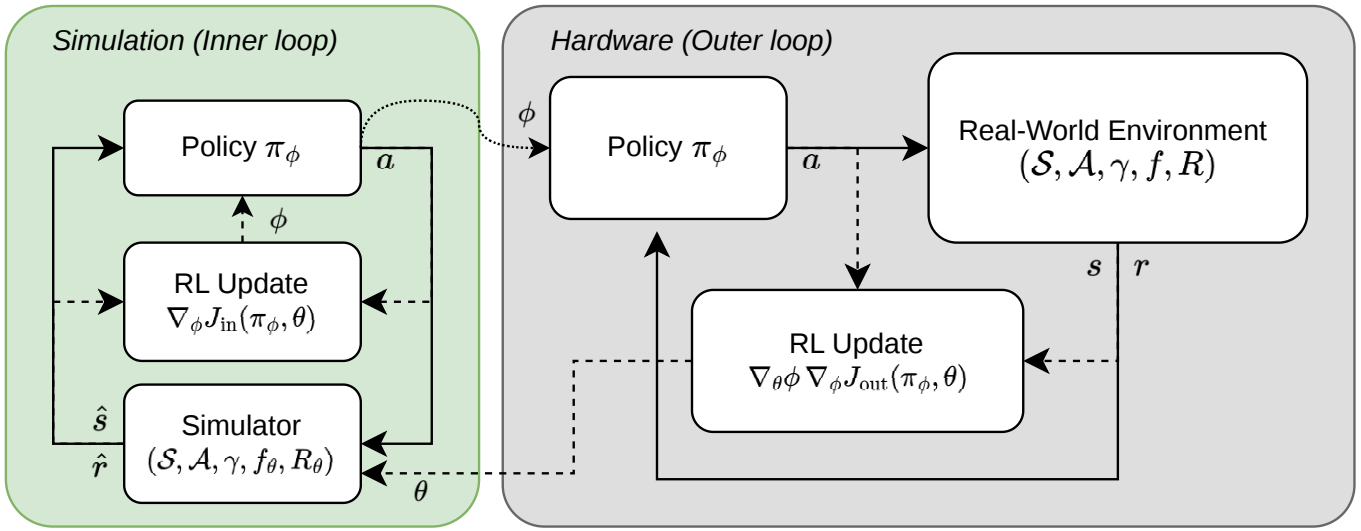


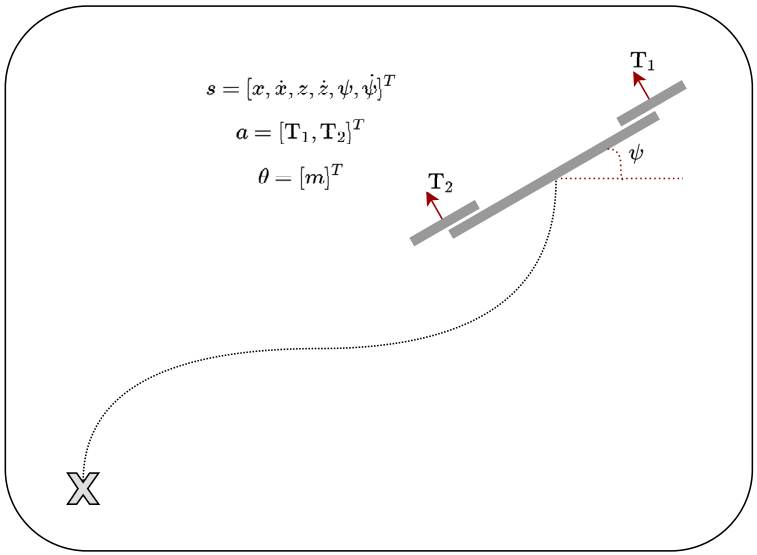
Example - In-Real RL over In-Sim RL

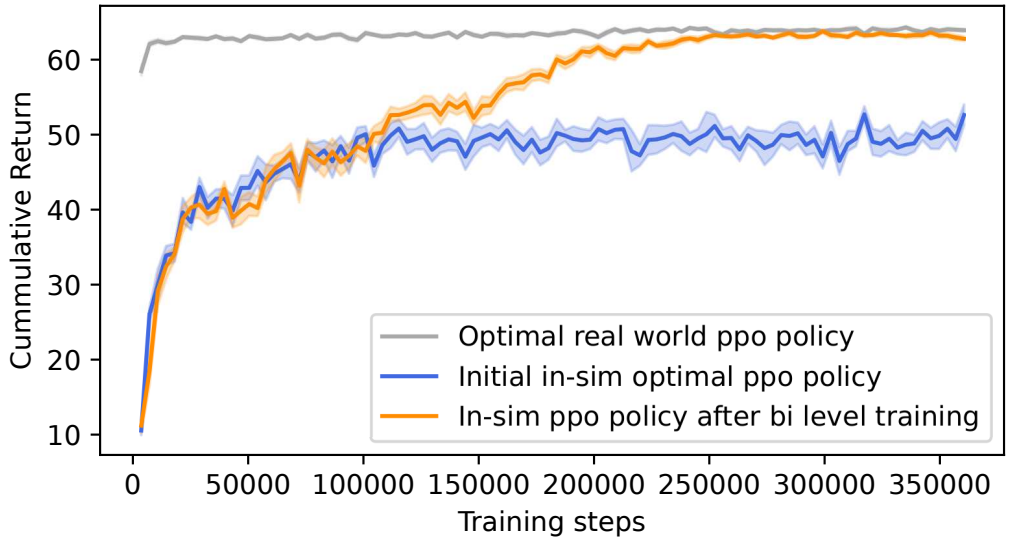




Example - In-Real RL over In-Sim RL







Norwegian Center on AI for Decisions

Email

- Start-up phase
- Recruiting soon!
- We can apply for funding to bring strong international postdocs



Thank you!